

第3章 分立元件基本电路

3.1 共发射极放大电路

3.2 共集电极放大电路

*3.3 共源极放大电路

3.4 分立元件组成的基本门电路



3.1 共发射极放大电路

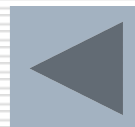
3.1.1 电路组成

3.1.2 静态分析

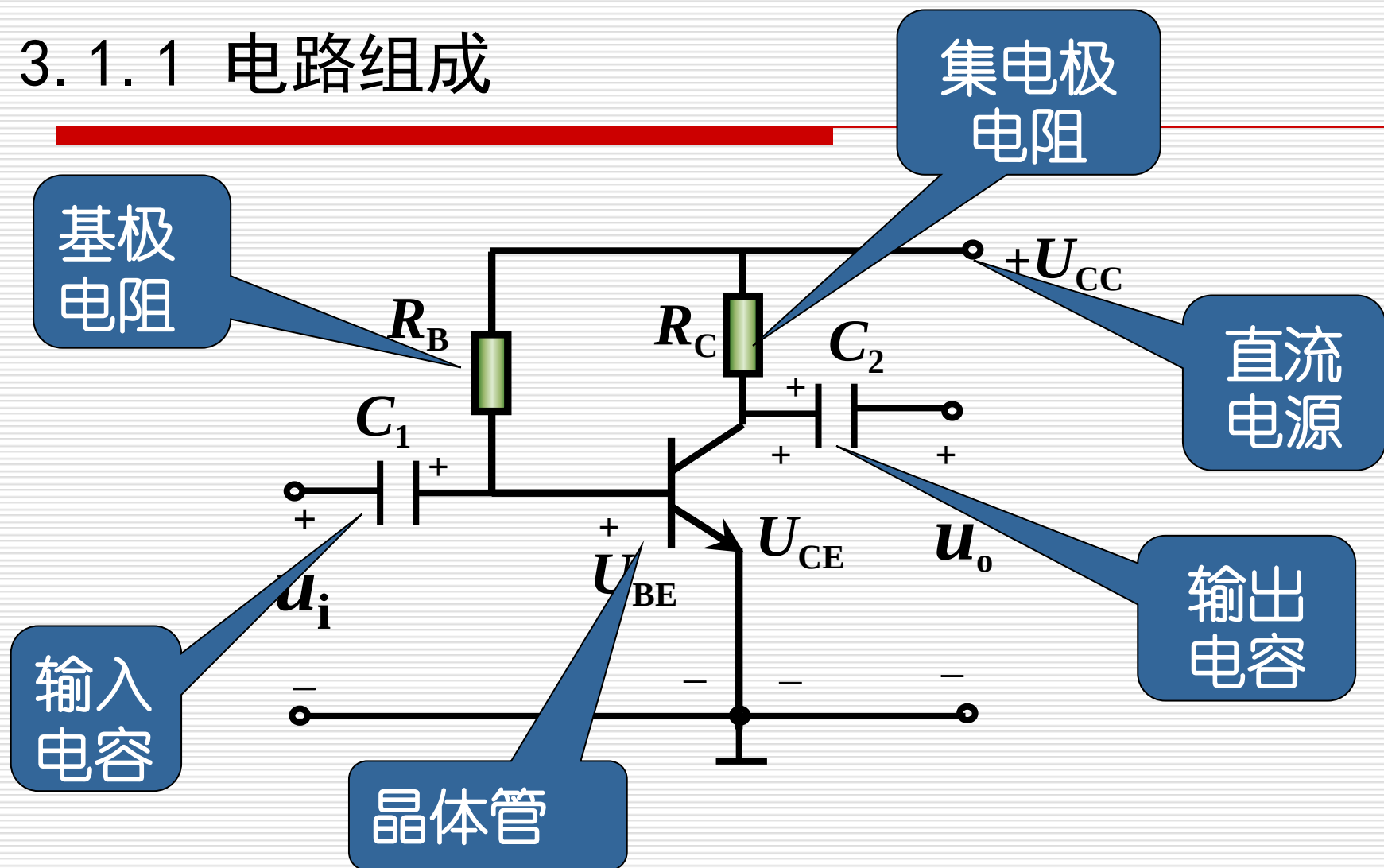
3.1.3 动态分析

3.1.4 静态工作点的稳定

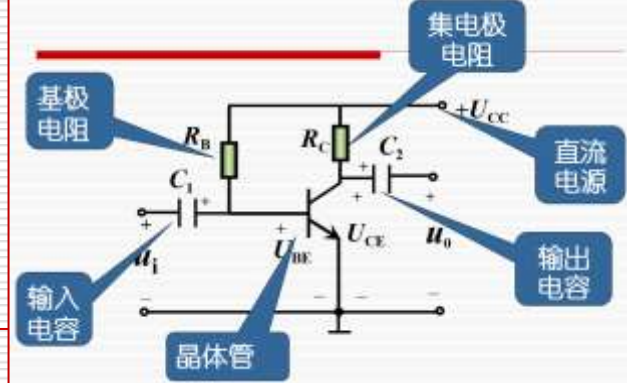
3.1.5 频率特性



3.1.1 电路组成



各元件的作用



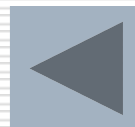
晶体管：电流放大作用，发射极正偏，集电极反偏；

直流电源 U_{CC} ：提供放大所需的能量；

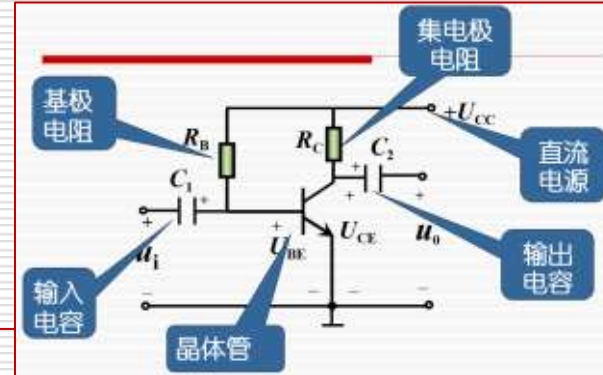
偏置电阻 R_B ：调节 R_B ，可调节基极电流；

集电极负载电阻 R_C ：将集电极电流的变化转换成电压的变化送到输出端；

耦合电容 C_1 、 C_2 ：隔直，不适用于低频电路。



3.1.2 静态分析



所谓静态是指当放大器没有输入信号 ($u_i = 0$) 时, 电路中各处的电压电流都是直流恒定值, 亦称为直流工作状态。

静态分析目的: 1) 确定三极管工作状态
2) 确定三极管静态工作点的合理性

静态分析内容: 在直流电源作用下, 确定三极管

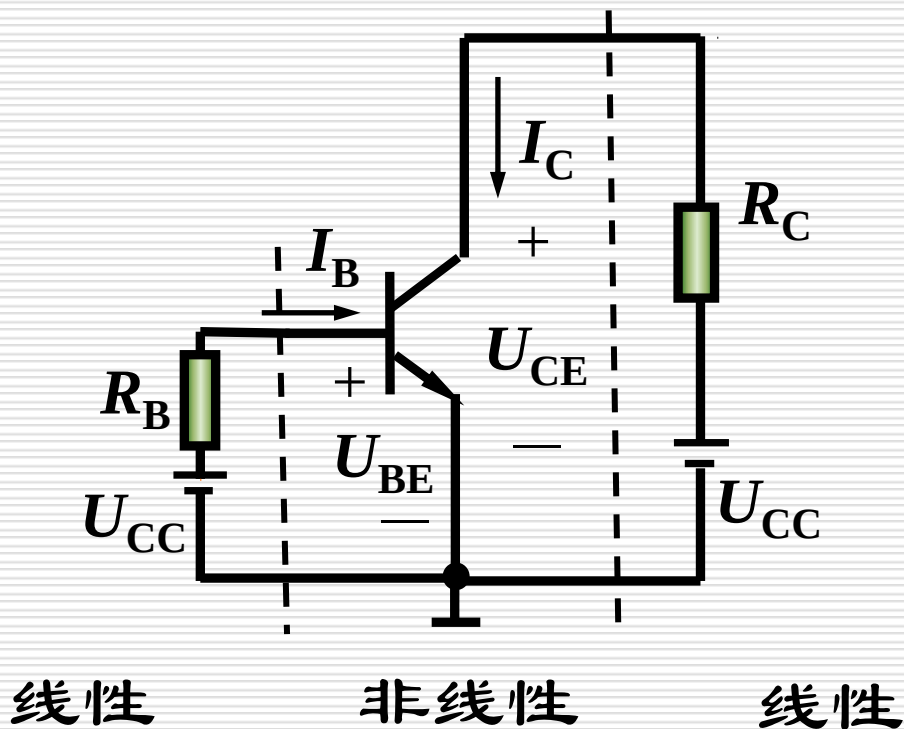
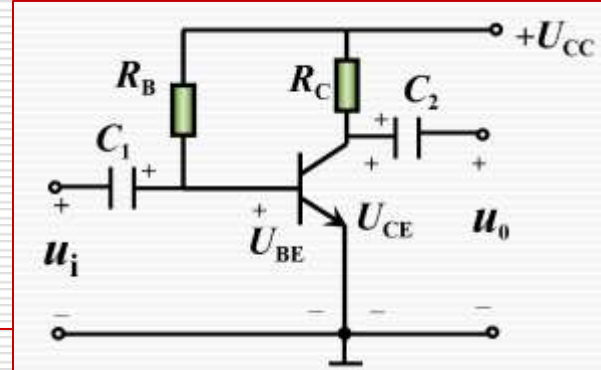
- 1) 基极电流 I_B ;
- 2) 集电极电流 I_C ;
- 3) 集电极与发射极之间的电压值 U_{CE} 。

静态分析方法: 1) 图解法;
2) 估算法。

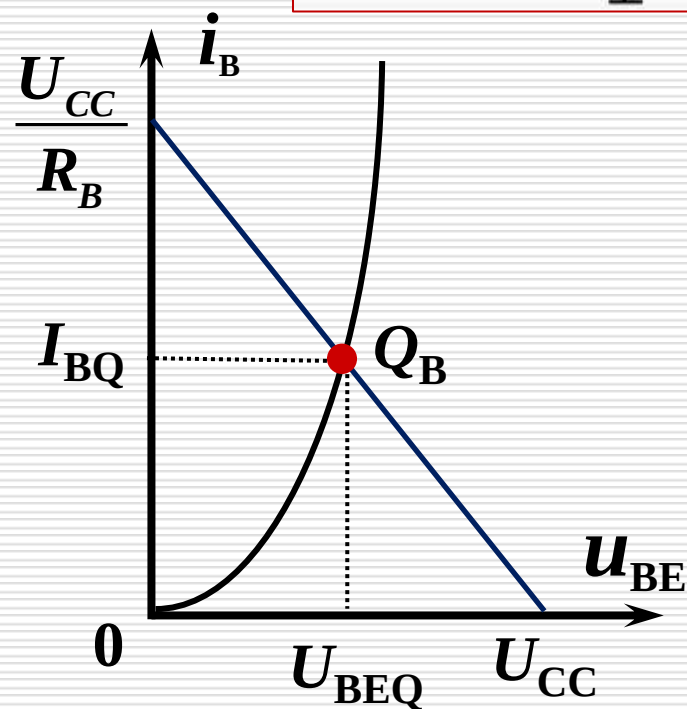


1) 图解法

输入回路



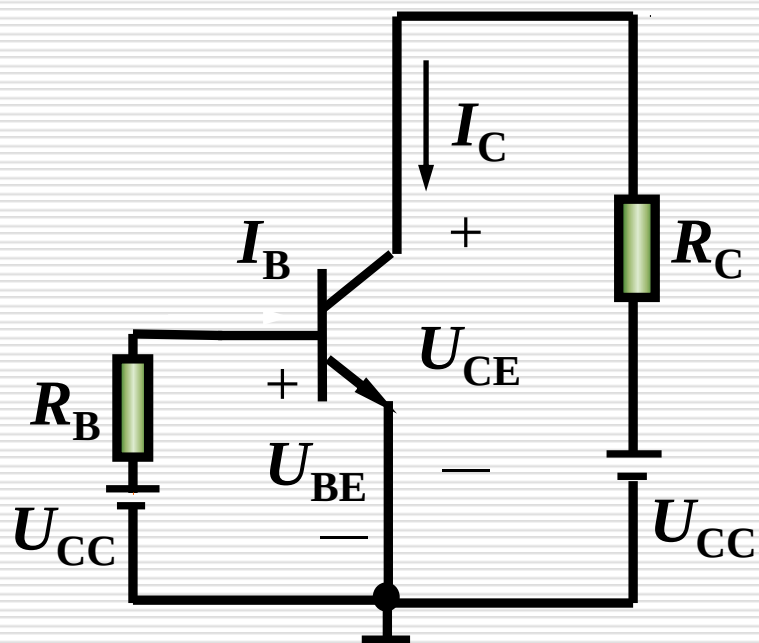
$$U_{BE} = U_{CC} - R_B I_B$$



$$i_B = f(u_{BE})$$

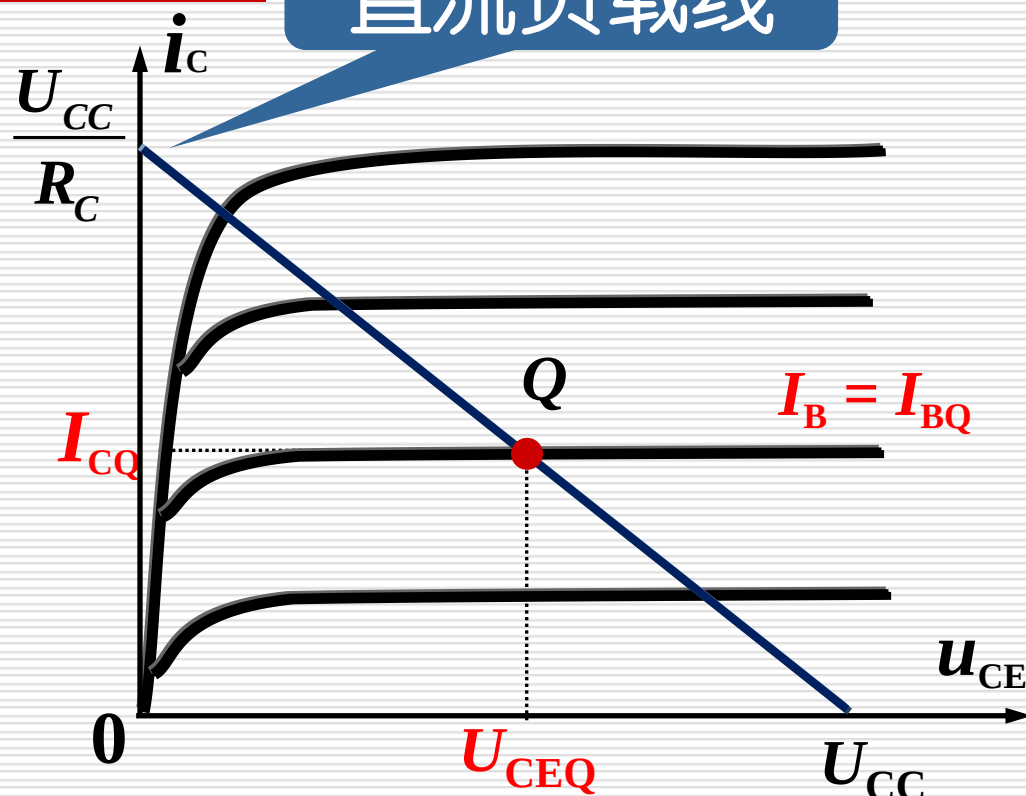


输出回路

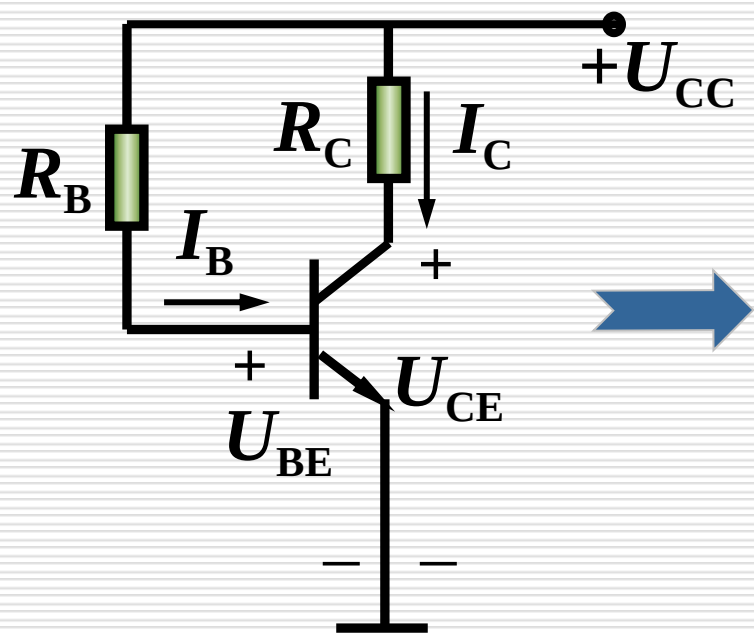
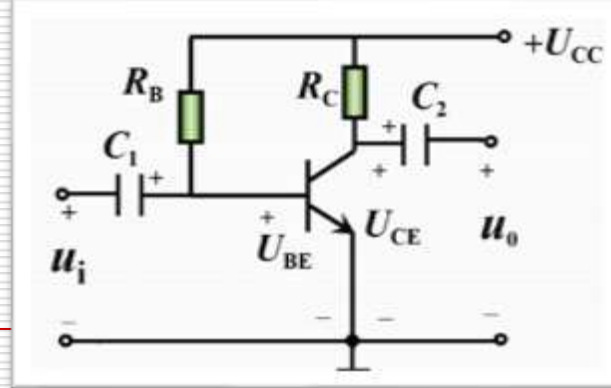


$$\begin{cases} I_C = f(U_{CE}) \mid I_B = \text{常量} \\ U_{CE} = U_{CC} - R_C I_C \end{cases}$$

直流负载线



2) 估算法

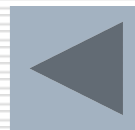


直流通路

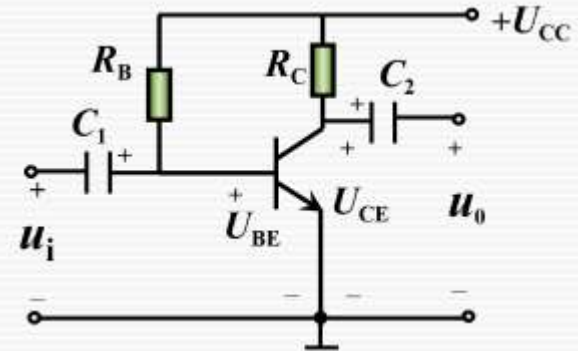
$$\begin{cases} U_{CC} = R_B I_B + U_{BE} \\ U_{CC} = U_{CE} + R_C I_C \end{cases}$$



$$\begin{cases} I_B = \frac{U_{CC} - U_{BE}}{R_B} \approx \frac{U_{CC}}{R_B} \\ I_C = \beta I_B \\ U_{CE} = U_{CC} - R_C I_C \end{cases}$$



3.1.3 动态分析

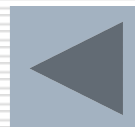


所谓动态是指放大电路有信号输入时，电路中各处的电压电流都处于变动工作状态。

动态分析内容：输入信号变化时，电路中各种变化量的变动情况和相互关系。

- 1) 电压放大倍数 A_U ；
- 2) 输入电阻 r_i ；
- 3) 输出电阻 r_o 。

动态分析方法：1) 图解法；
2) 微变等效电路分析方法。

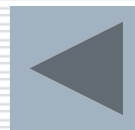
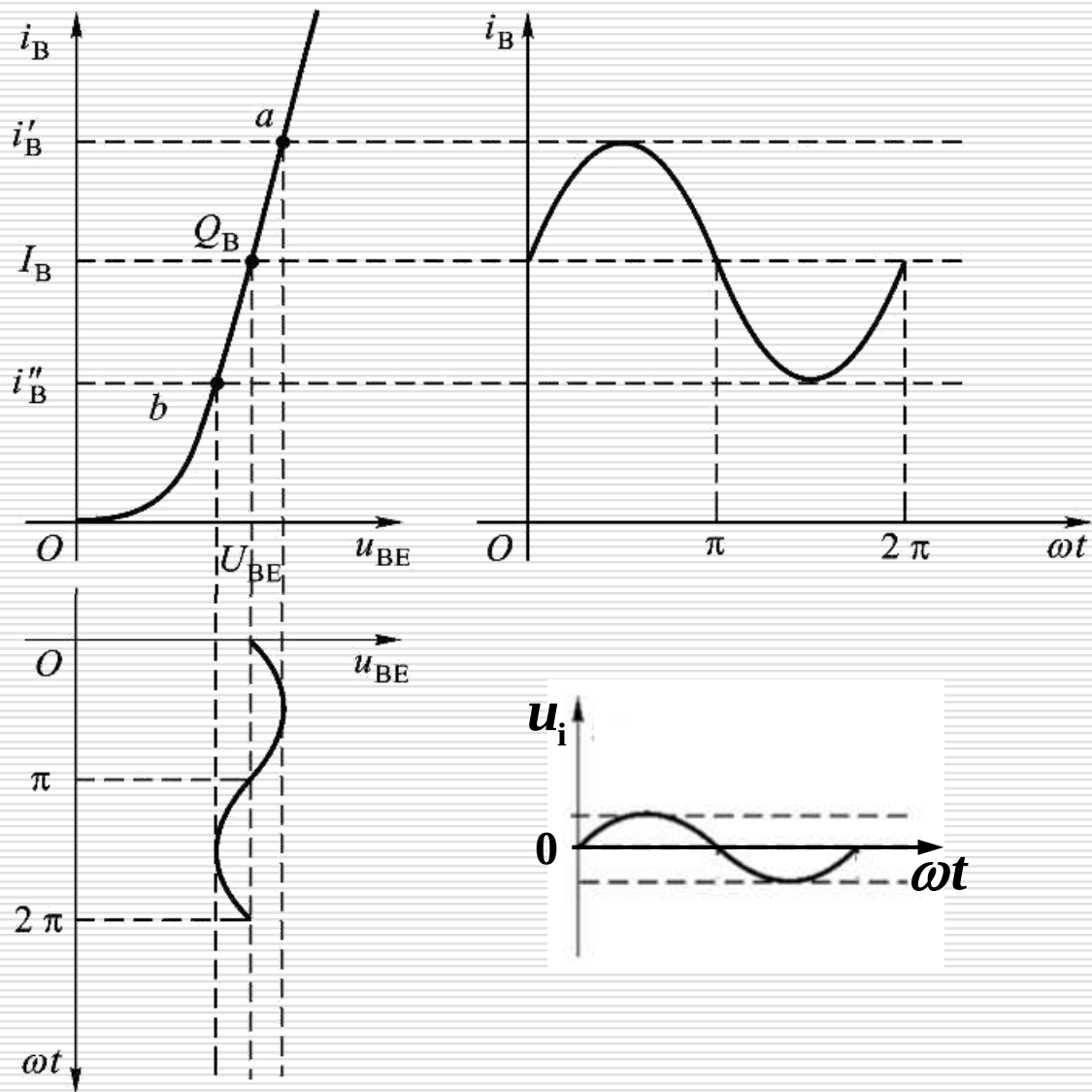


1) 图解法

输入电路

$$u_{BE} = U_{BE} + u_i$$

$$i_B = I_B + i_b$$



输出电路

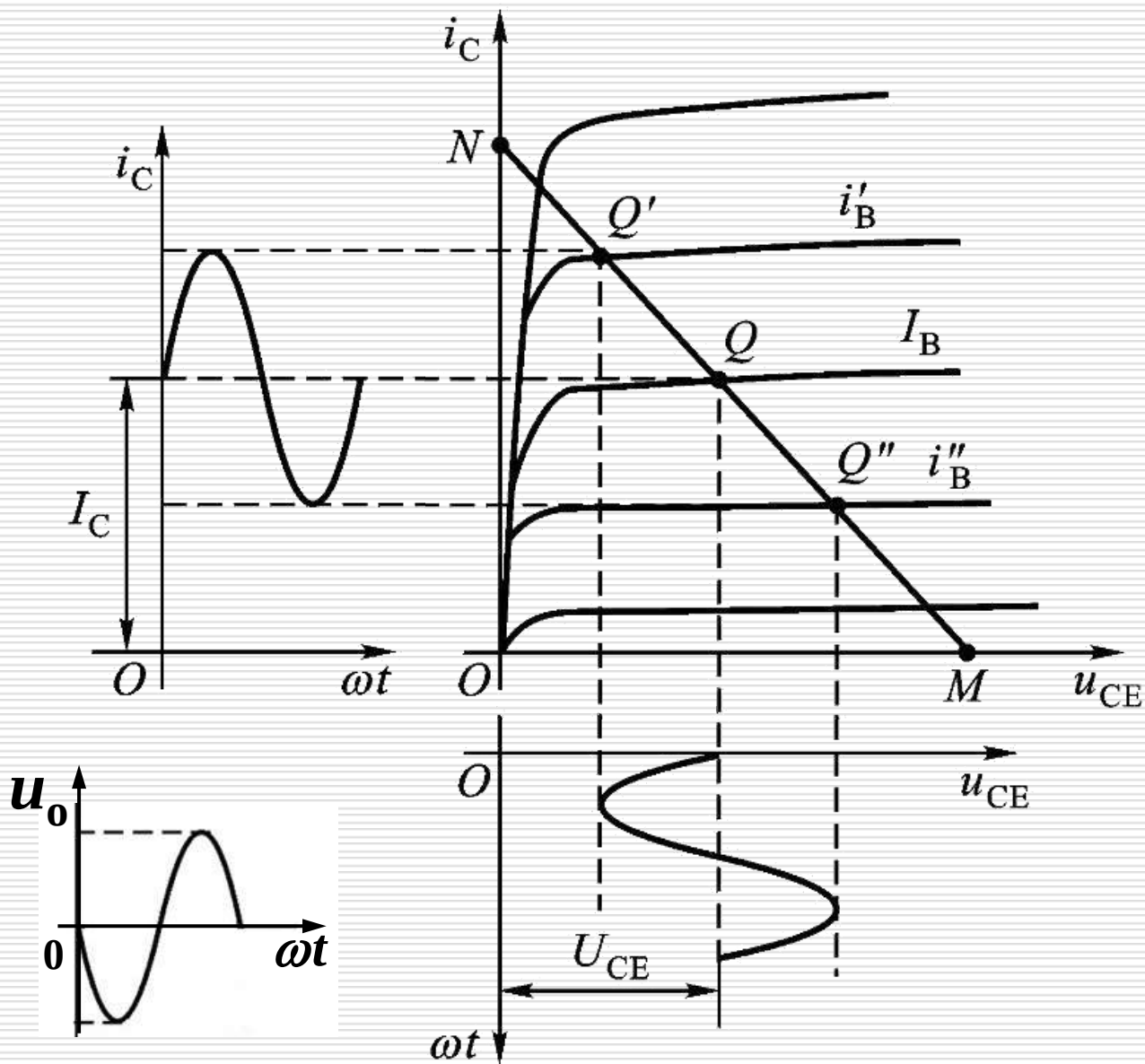
$$i_C = I_C + i_c$$

$$u_{CE} = U_{CE} + u_{ce}$$

经电容隔直后：

$$u_o = u_{CE} - U_{CE}$$

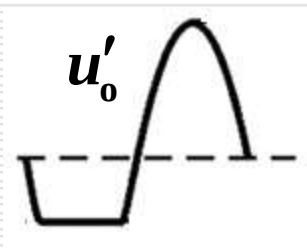
$$= u_{ce}$$



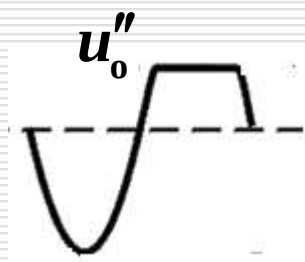
工作点与
波形失真

Q点过高
饱和失真

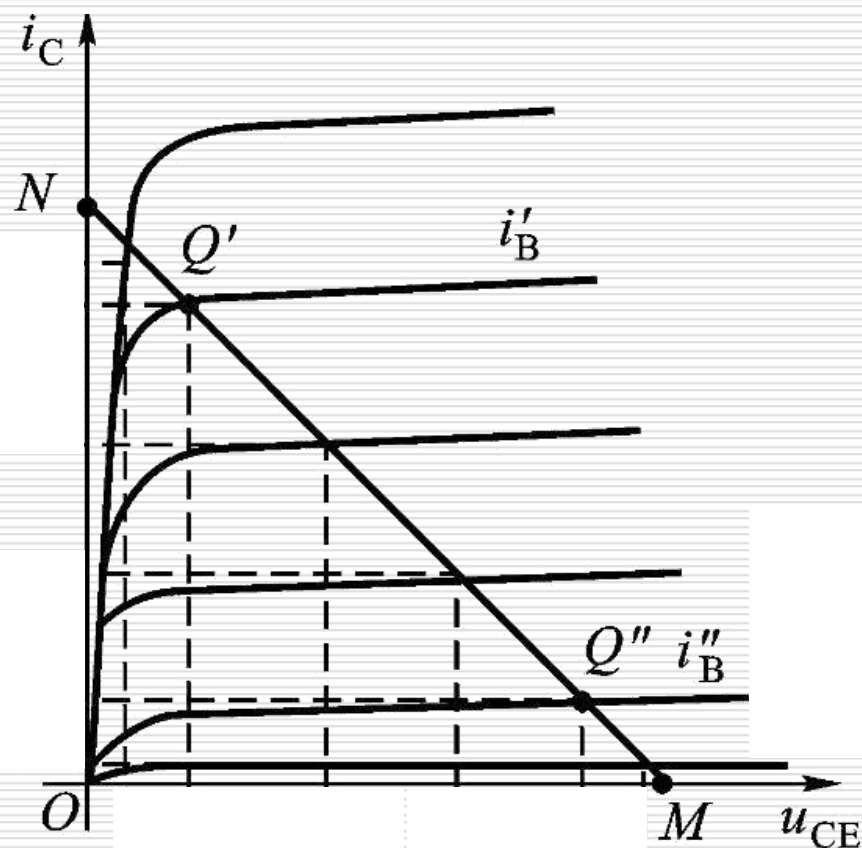
Q点过低
截止失真



饱和失真



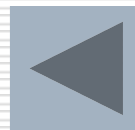
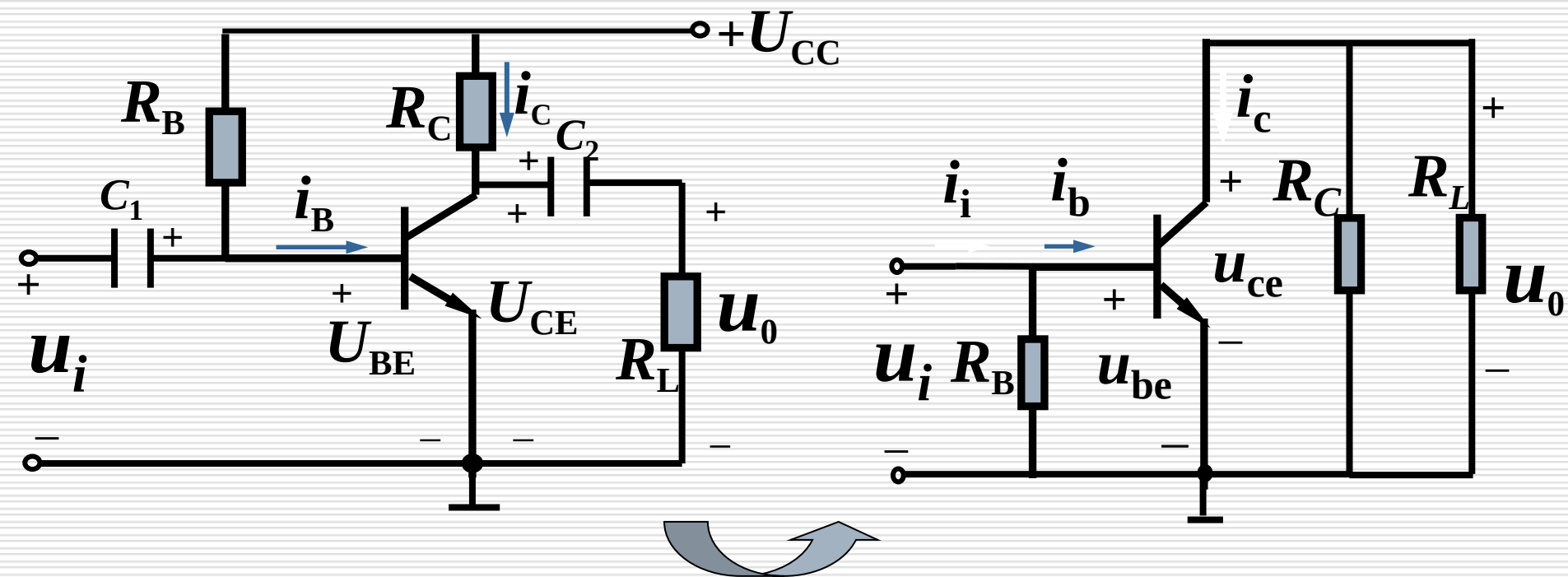
截止失真



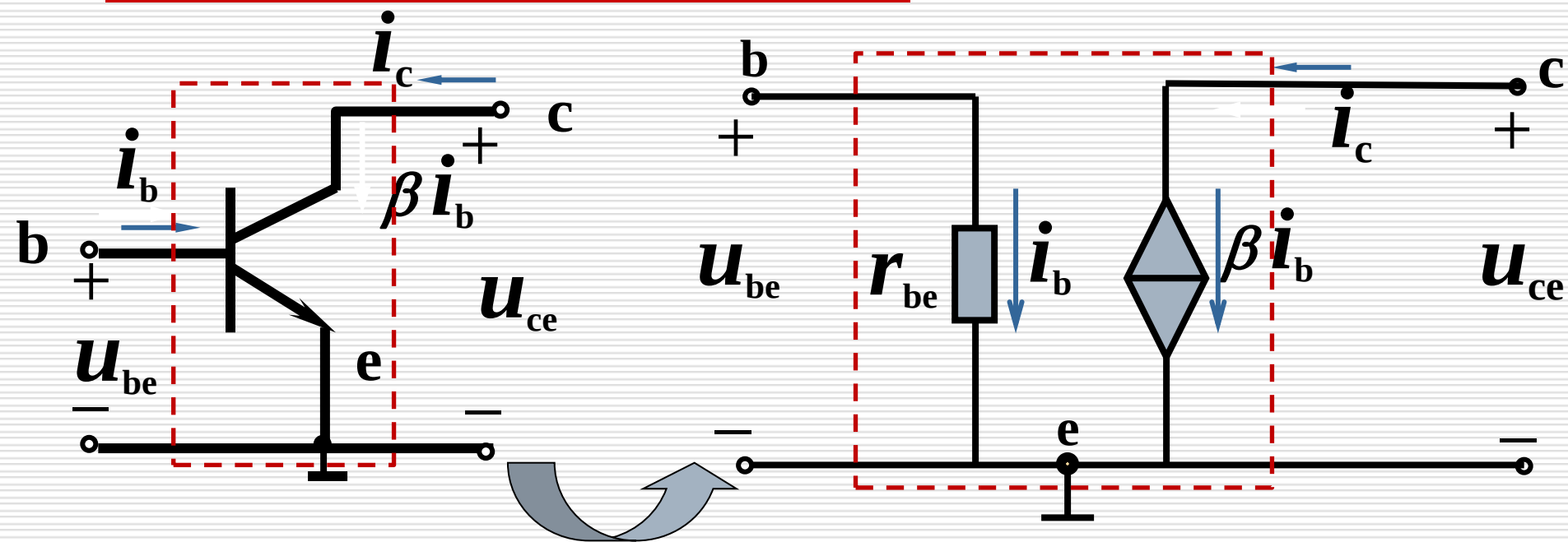
2) 微变等效电路分析法

交流通路

交流通路：将电路中的电容和直流电源短路。



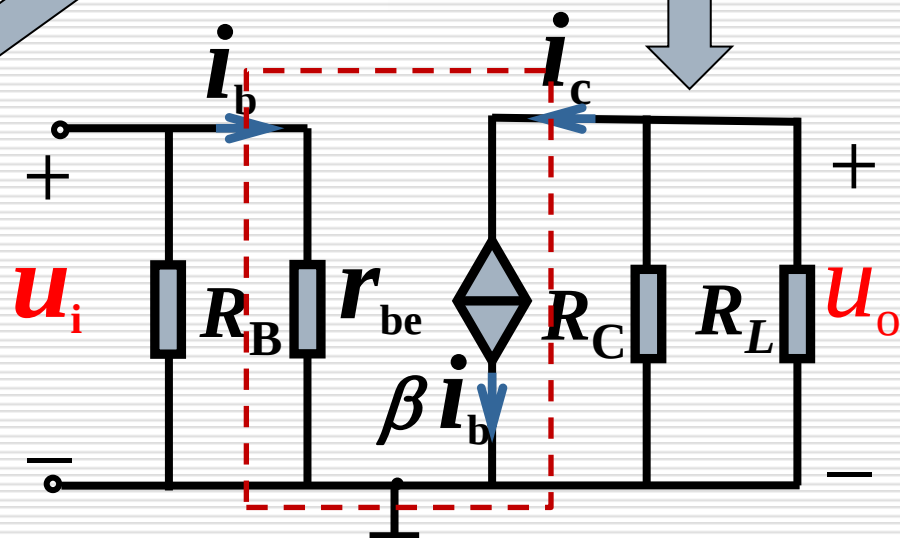
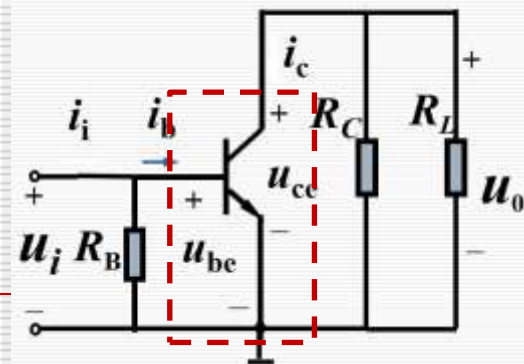
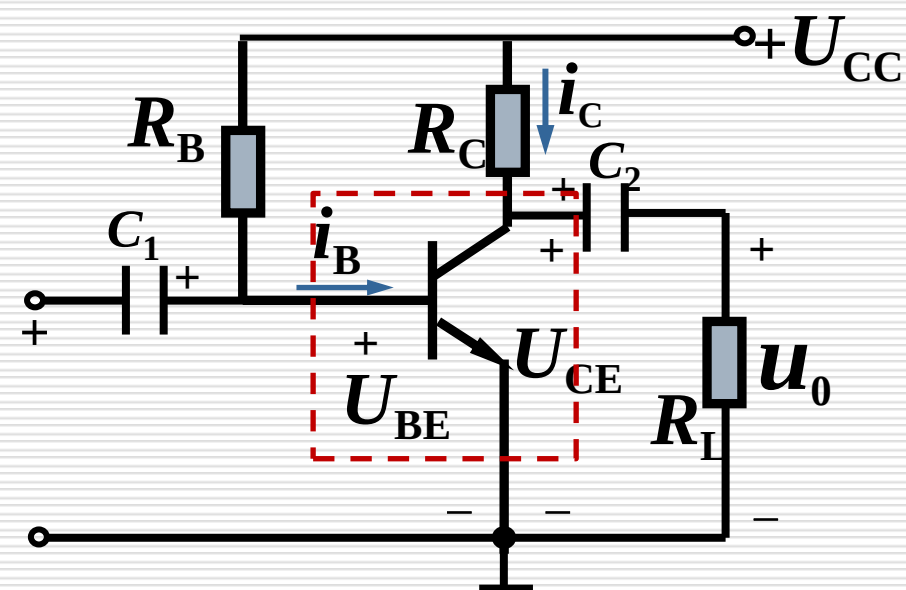
晶体管的小信号模型 (P23-P24 (5))



微变等效电路：将晶体管用小信号模型来代替。



微变等效电路



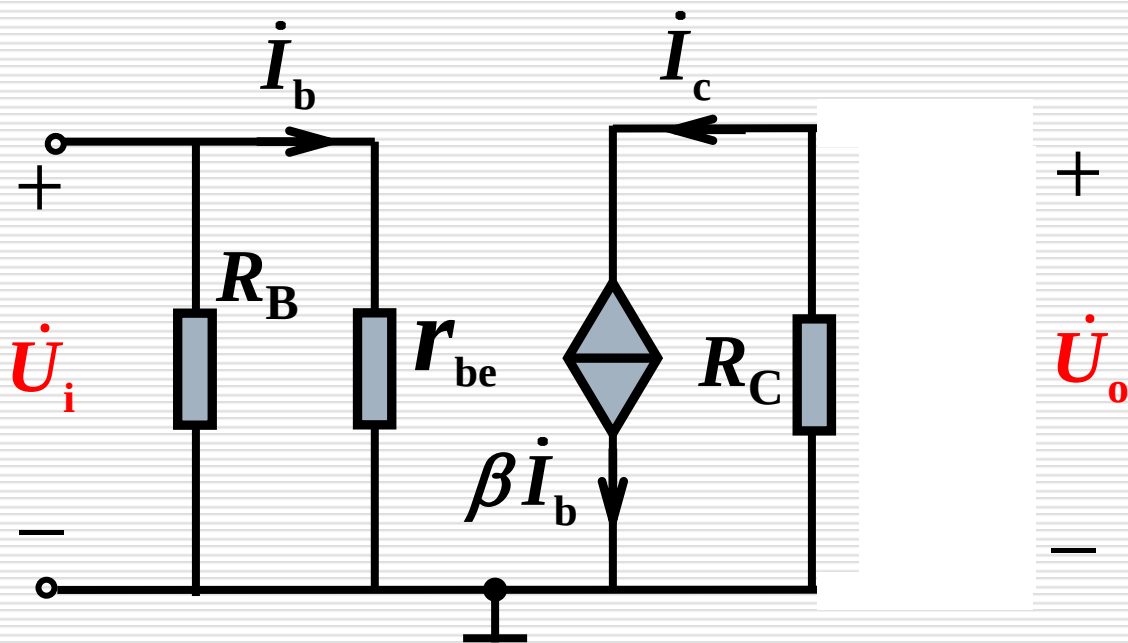
(1) 电压放大倍数

不带负载时

$$\dot{U}_i = r_{be} \dot{I}_b$$

$$\dot{U}_o = -\beta R_C \dot{I}_b$$

$$A_u = \frac{\dot{U}_o}{\dot{U}_i} = -\beta \frac{R_C}{r_{be}}$$

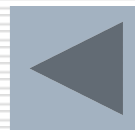


上式说明，选择适当的参数，可使信号得到放大。

A_u 为负值，说明输出信号与输入信号反相，即共射极放大电路具有倒相作用。

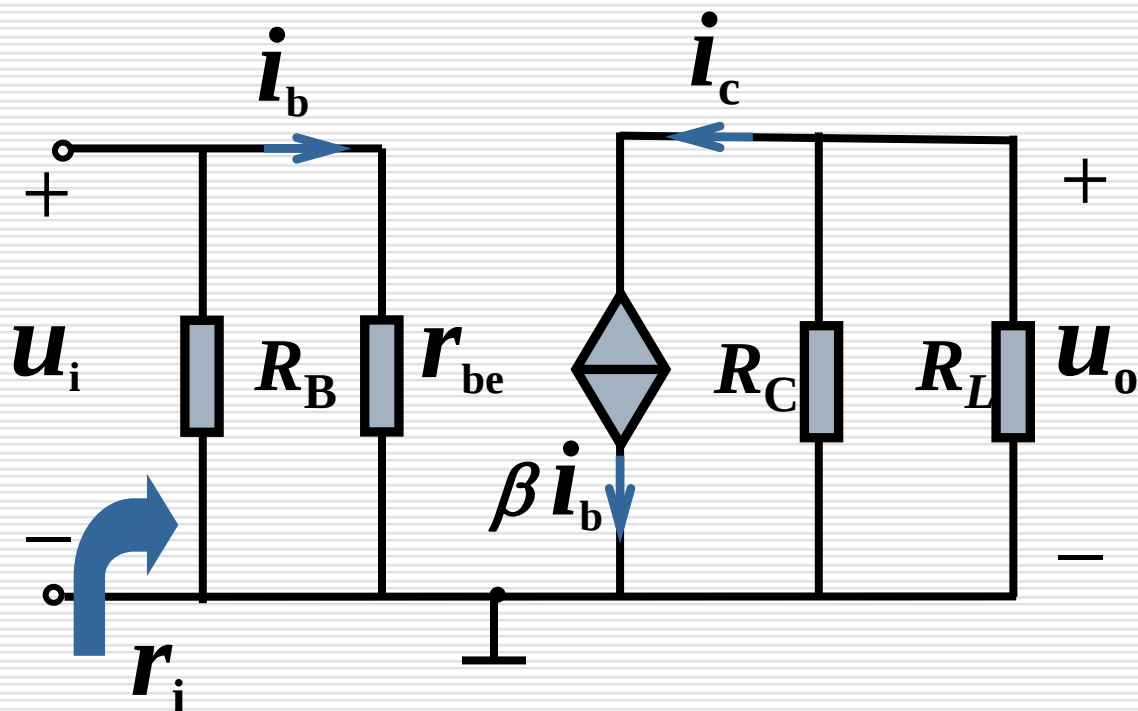
带负载时
$$A_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{-\beta \dot{I}_b (R_C // R_L)}{r_{be} \dot{I}_b} = -\beta \frac{(R_C // R_L)}{r_{be}}$$

可见带负载使放大倍数下降。



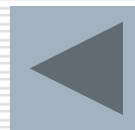
(2) 输入电阻

对信号源而言，放大电路相当于它的负载，负载电阻即为放大电路的输入电阻。



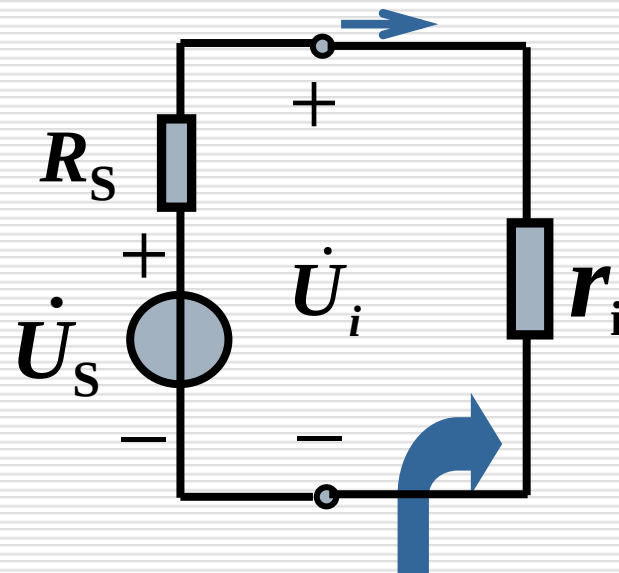
放大电路的输入电阻：
$$r_i = \frac{\dot{U}_i}{\dot{I}_i}$$

根据以上的微变等效电路：
$$r_i = R_B // r_{be}$$



如右图所示，把一个内阻为 R_S ，源电压为 $\dot{U}_S$ 的信号源加到放大电路的输入端，由于 r_i 的存在，实际的 $\dot{U}_i$ 为：

$$\dot{U}_i = \frac{r_i}{R_S + r_i} \dot{U}_S$$

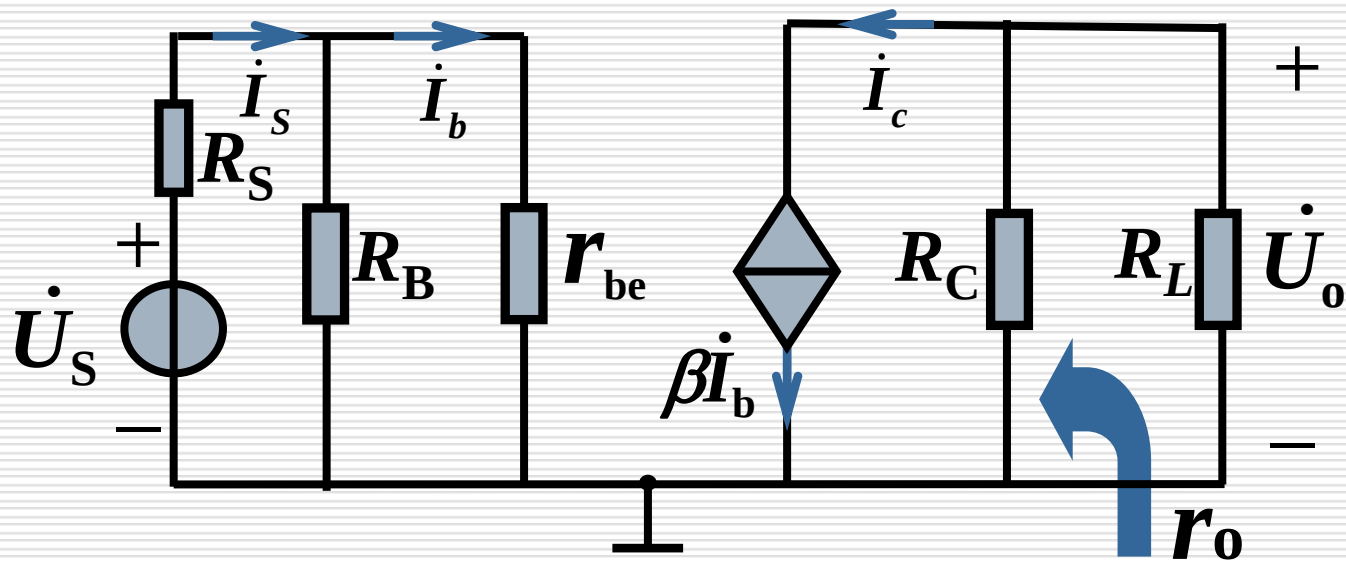


r_i 是衡量放大电路对输入电压衰减程度的重要指标。



(3) 输出电阻

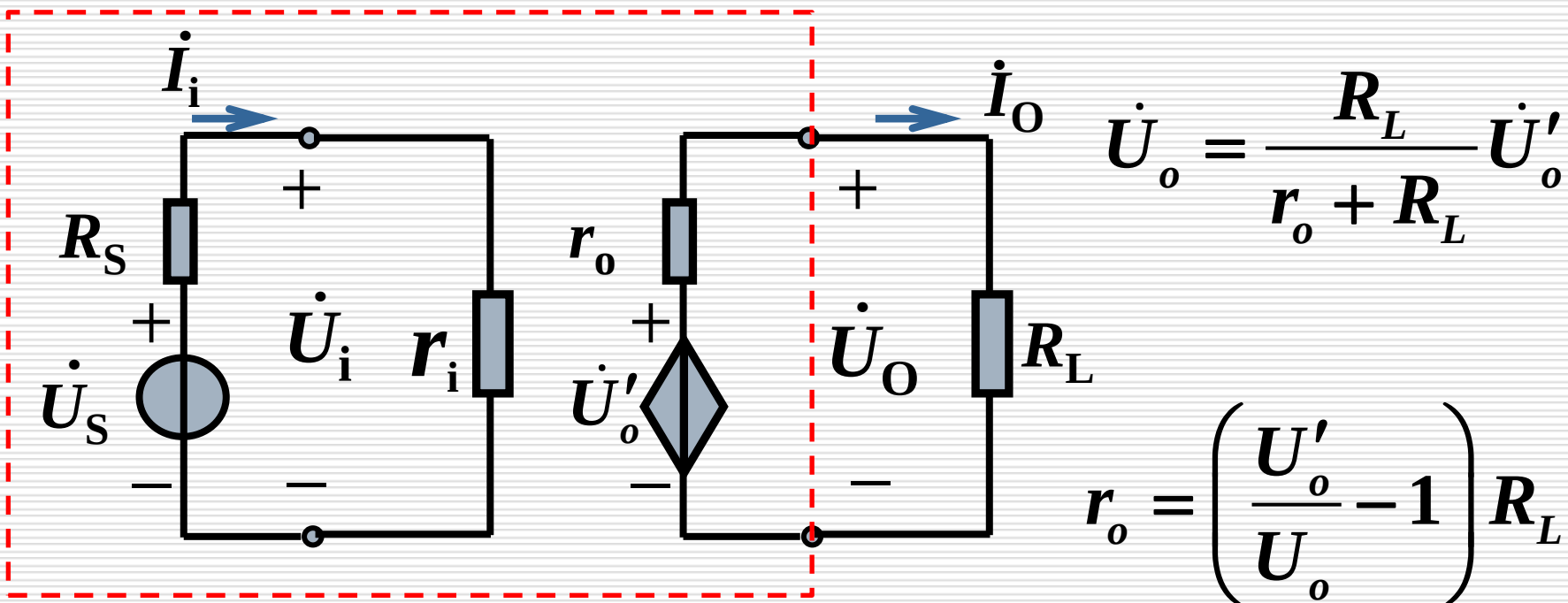
对负载而言, 放大电路相当于一个具有内阻的信号源, 信号源的内阻就是放大电路的输出电阻。



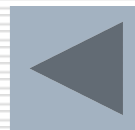
$$r_o = \frac{\dot{U}}{\dot{I}} = R_C$$

实验法求输出电阻

分别测放大器空载电压 $\dot{U}'_o$ 及已知 R_L 的负载电压 $\dot{U}_o$ 。

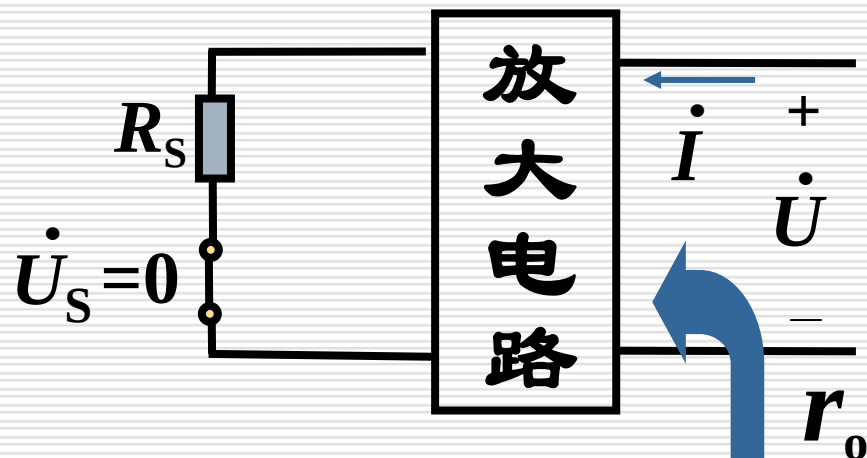


放大电路



外施电压法输出电阻

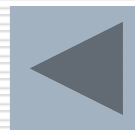
$$\therefore r_o = \frac{\dot{U}}{\dot{I}}$$



可用外加电压法求 r_o

对于前面所述的共发射极放大电路

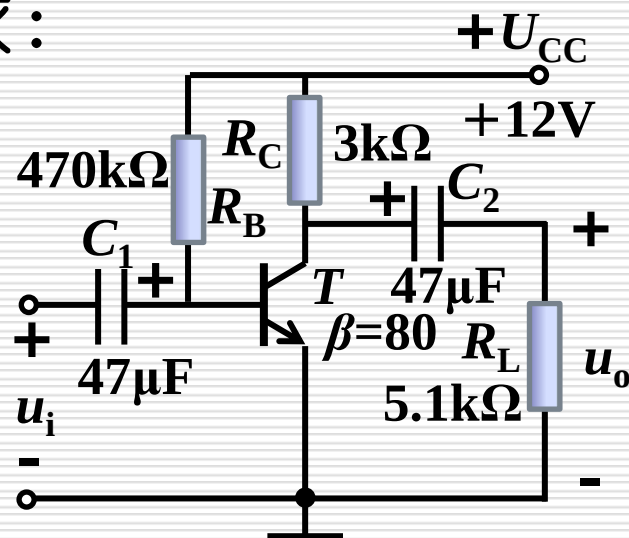
$$\therefore r_o = \frac{\dot{U}}{\dot{I}} = R_C$$



[例题3.1.1] 共射极放大电路如图，设 u_i 为正弦信号
已知电路参数， $U_{BE}=0.7V$ 。求：

- (1) 不接负载时的放大倍数
- (2) 接负载 R_L 时的放大倍数
- (3) 放大电路的 r_i 、 r_o

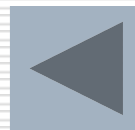
[解] 先进行静态分析



$$I_B = \frac{U_{CC} - U_{BE}}{R_B} = \frac{12 - 0.7}{470 \times 10^3} = 0.024\text{mA} \quad I_C = 1.92\text{mA}$$

$$I_E = (1 + \beta) I_B = 81 \times 0.024 = 1.94\text{mA} \quad U_{CE} = 6.24\text{V}$$

$$r_{be} = 200 + (1 + \beta) \frac{26}{I_E} = 200 + 81 \times \frac{26}{1.94} = 1.286\text{k}\Omega$$



(1) 不接负载 R_L 时的电压放大倍数

$$A_{uo} = -\beta \frac{R_C}{r_{be}} = -80 \times \frac{3 \times 10^3}{1.286 \times 10^3} = -186.6$$

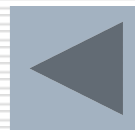
(2) 接负载 R_L 时的放大倍数

$$A_u = -\beta \frac{R'_L}{r_{be}} = -80 \times \frac{1.89 \times 10^3}{1.286 \times 10^3} = -117.6$$

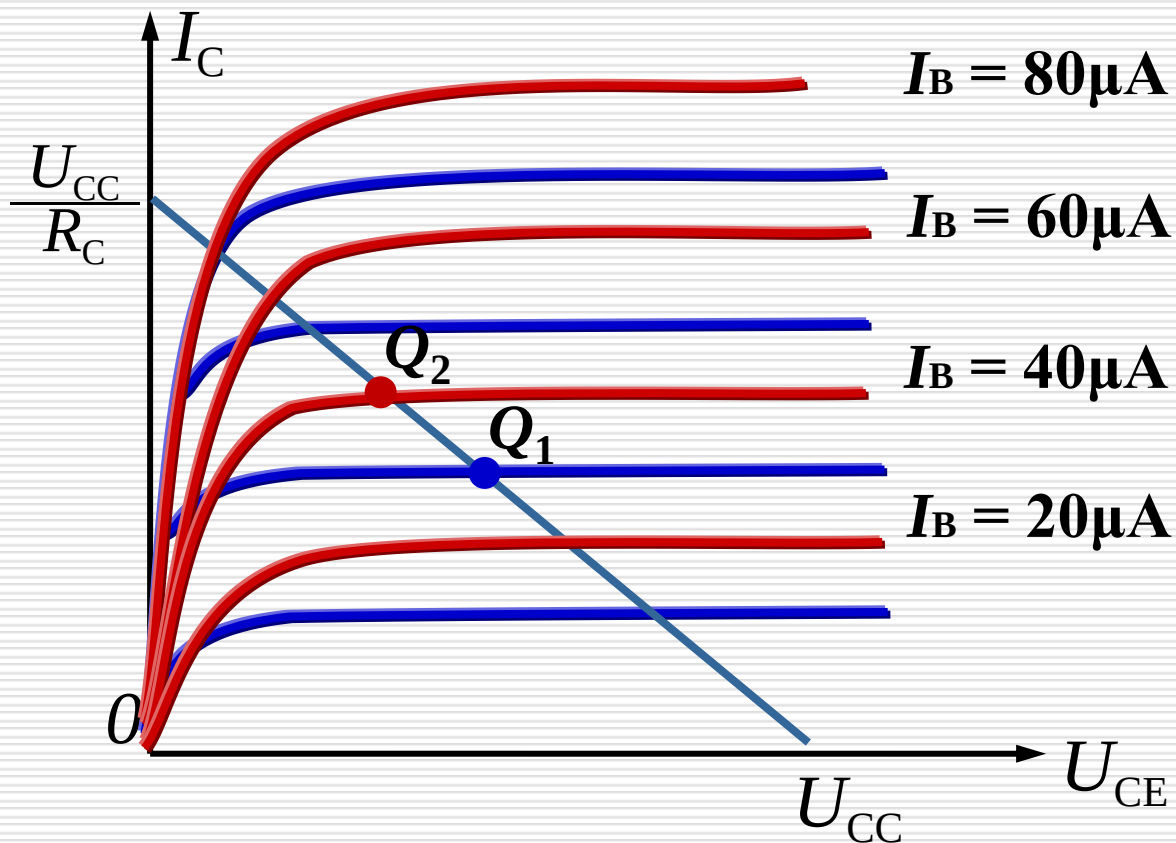
其中
$$R'_L = \frac{R_C R_L}{R_C + R_L} = \frac{3 \times 5.1}{3 + 5.1} = 1.89 \text{k}\Omega$$

(3) 放大电路的输入电阻 r_i 、输出电阻 r_o

$$r_i = \frac{R_B r_{be}}{R_B + r_{be}} = \frac{470 \times 1.286}{470 + 1.286} = 1.28 \text{k}\Omega \quad r_o = R_C = 3 \text{k}\Omega$$



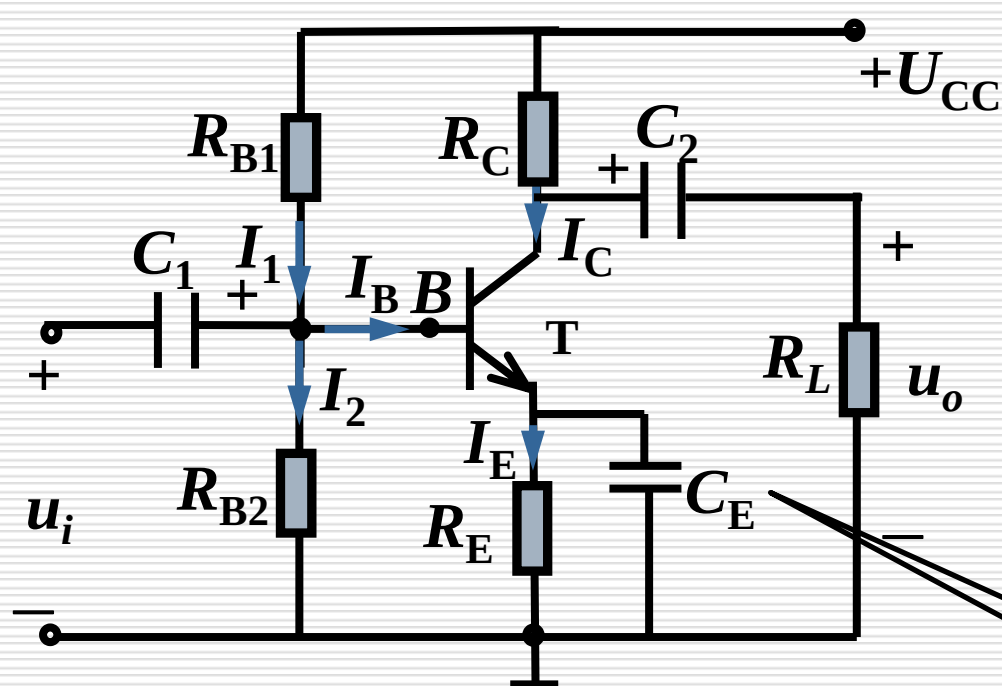
3.1.4 静态工作点的稳定



温度升高时，静态工作点将沿直流负载线上移。



静态工作点稳定的放大电路



$$U_B \approx \frac{R_{B2}}{R_{B1} + R_{B2}} U_{CC}$$

$$I_E = \frac{U_E}{R_E} = \frac{U_B - U_{BE}}{R_E}$$

硅管:

$$I_1 = (5 \sim 10) I_E, U_B = 3 \sim 5V$$

锗管:

$$I_1 = (10 \sim 20) I_E, U_B = 1 \sim 3V$$

静态工作点稳定过程:

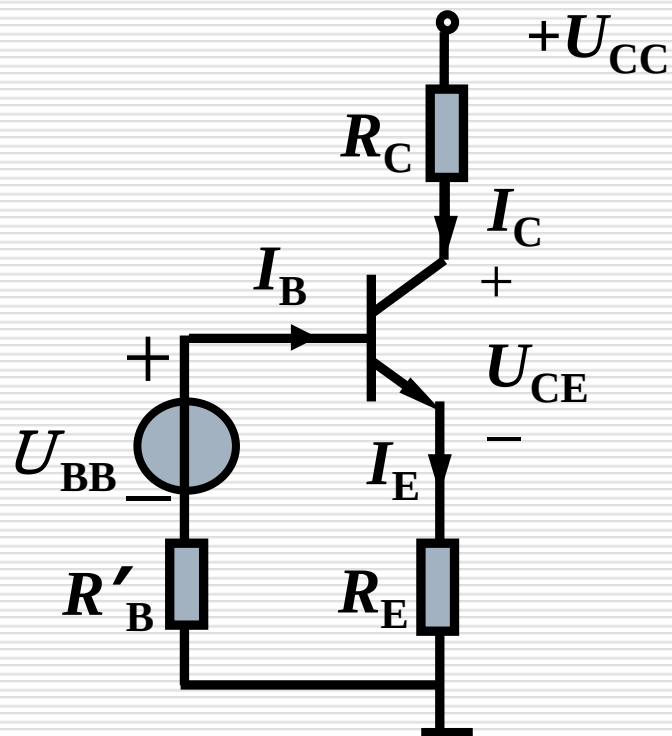
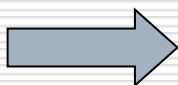
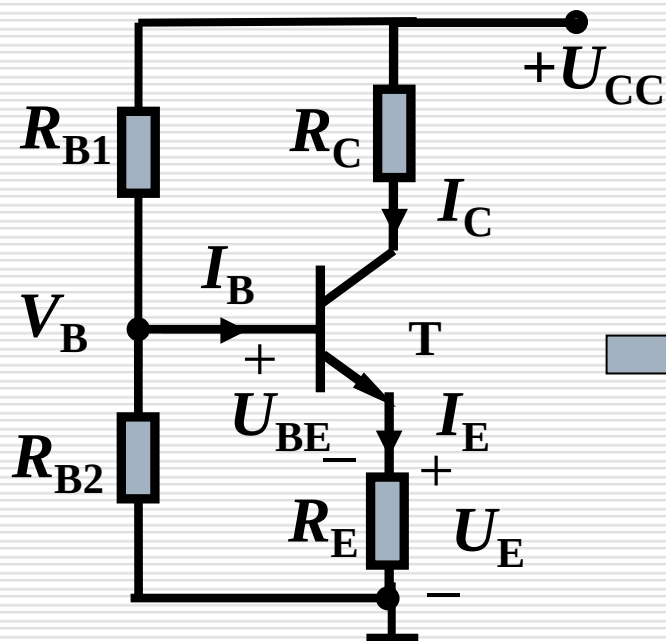
设 $\beta \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow \rightarrow U_E \uparrow \rightarrow U_B$ 基本不变

$\uparrow \leftarrow I_C \downarrow \leftarrow I_B \downarrow \leftarrow U_{BE} \downarrow$

C_E : 旁路电容



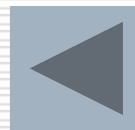
静态分析



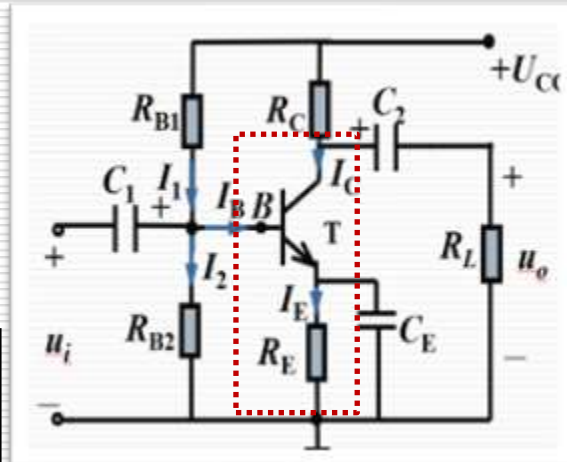
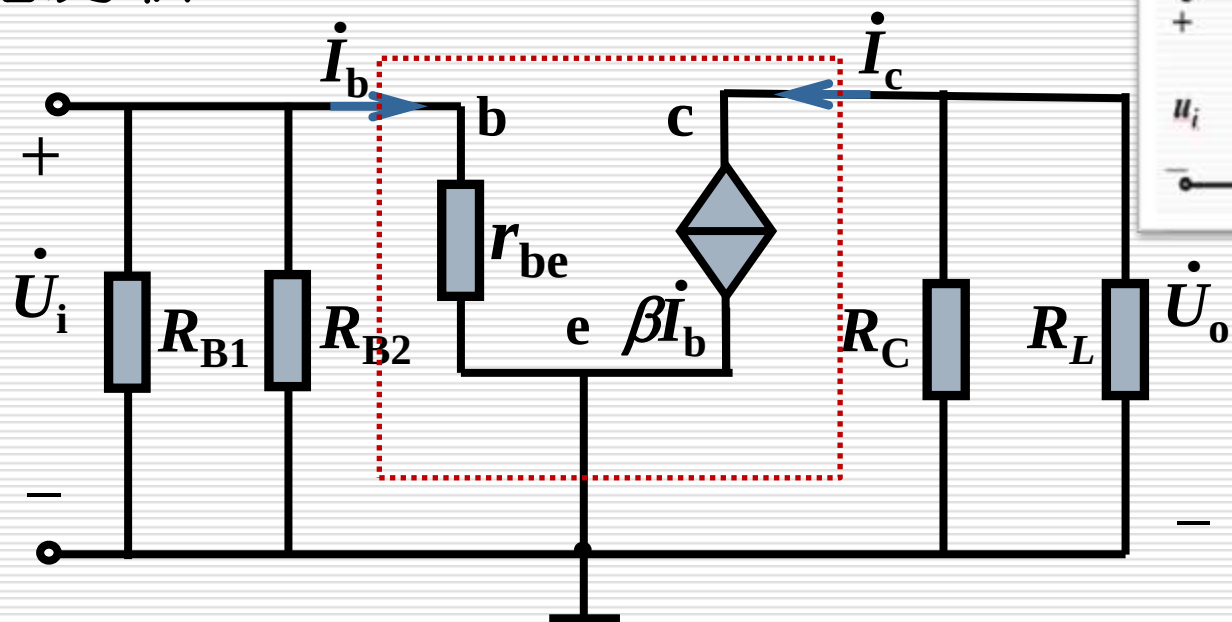
$$U_{BB} = \frac{R_{B2}}{R_{B1} + R_{B2}} U_{CC}$$

$$R'_B = \frac{R_{B1} R_{B2}}{R_{B1} + R_{B2}}$$

$$I_B = \frac{U_{BB} - U_{BE}}{R'_B + (1 + \beta) R_E}$$



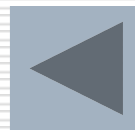
动态分析



$$A_u = -\beta \frac{R_C // R_L}{r_{be}}$$

$$r_i = R_{B1} // R_{B2} // r_{be}$$

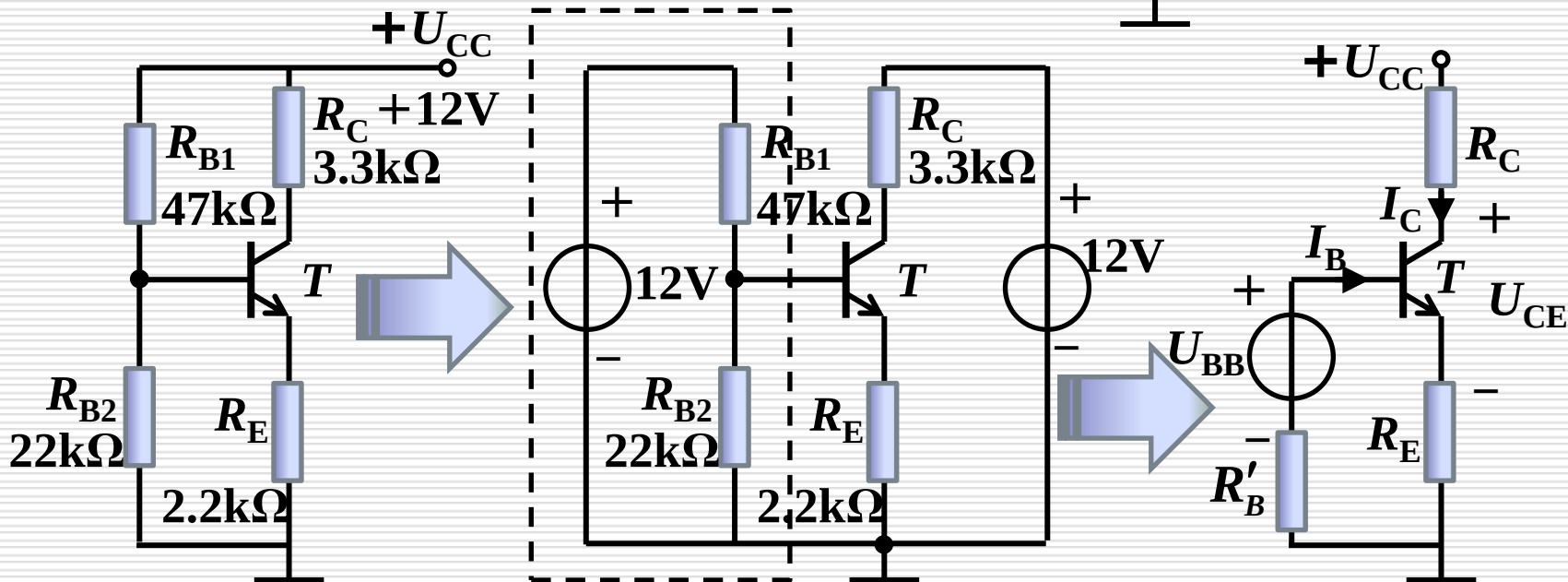
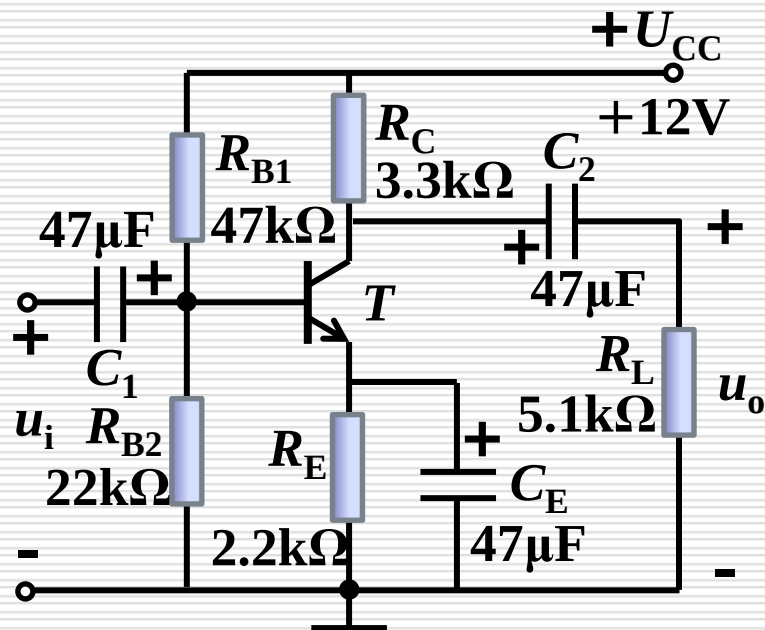
$$r_o = R_C$$



[例题3.1.2] 已知电路参数,
 $U_{BE}=0.7V$, $\beta=80$ 。求:

- (1) 电路的静态工作点
- (2) 电压放大倍数
- (3) 电路的 r_i 、 r_o

[解] (1) 求静态工作点



$$U_{BB} = \frac{R_{B2}}{R_{B1} + R_{B2}} U_{CC} = \frac{22 \times 10^3}{(47 + 22) \times 10^3} \times 12 = 3.83\text{V}$$

$$R'_B = \frac{R_{B1} R_{B2}}{R_{B1} + R_{B2}} = \frac{47 \times 22}{47 + 22} = 14.99\text{k}\Omega$$

输入回路方程 $U_{BB} = U_{BE} + R'_B I_B + (1 + \beta) R_E I_B$

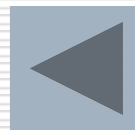
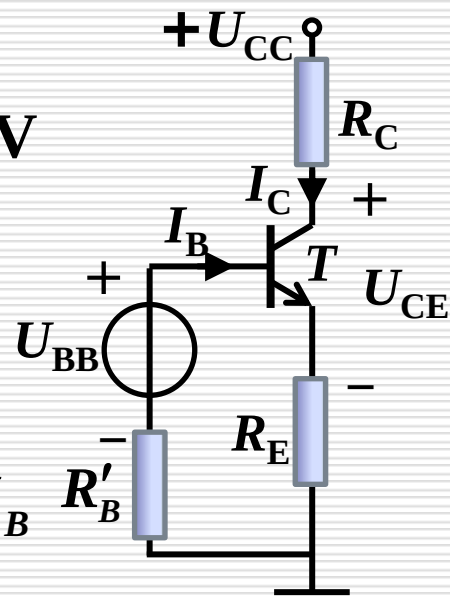
$$\text{故 } I_B = \frac{U_{BB} - U_{BE}}{R'_B + (1 + \beta) R_E} = \frac{3.83 - 0.7}{(14.99 + 81 \times 2.2) \times 10^3} = 0.0162\text{mA}$$

$$I_C = \beta I_B = 80 \times 0.0162 = 1.30\text{mA}$$

$$I_E = \beta I_B = 81 \times 0.0162 = 1.31\text{mA}$$

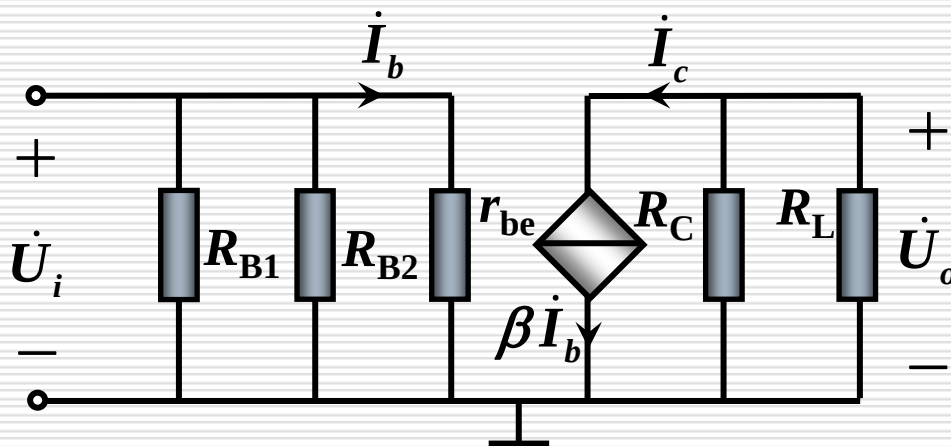
$$U_{CE} = U_{CC} - R_C I_C - R_E I_E$$

$$\begin{aligned} &= 12 - 3.3 \times 10^3 \times 1.3 \times 10^{-3} - 2.2 \times 10^3 \times 1.31 \times 10^{-3} \\ &= 4.83\text{V} \end{aligned}$$



(2) 电压放大倍数

由于旁路电容 C_E 的作用, R_E 被交流短路, 晶体管的 E 极直接接地



等效负载电阻

$$R'_L = \frac{R_C R_L}{R_C + R_L} = \frac{3.3 \times 5.1}{3.3 + 5.1} = 2\text{k}\Omega$$

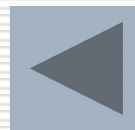
电压放大倍数

$$A_u = -\beta \frac{R'_L}{r_{be}} = -80 \times \frac{2}{1.81} = -88.4$$

(3) 电路的 r_i 、 r_o

$$r_i = R_{B1} // R_{B2} // r_{be} = 1.61\text{k}\Omega$$

$$r_o = R_C = 3.3\text{k}\Omega$$



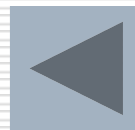
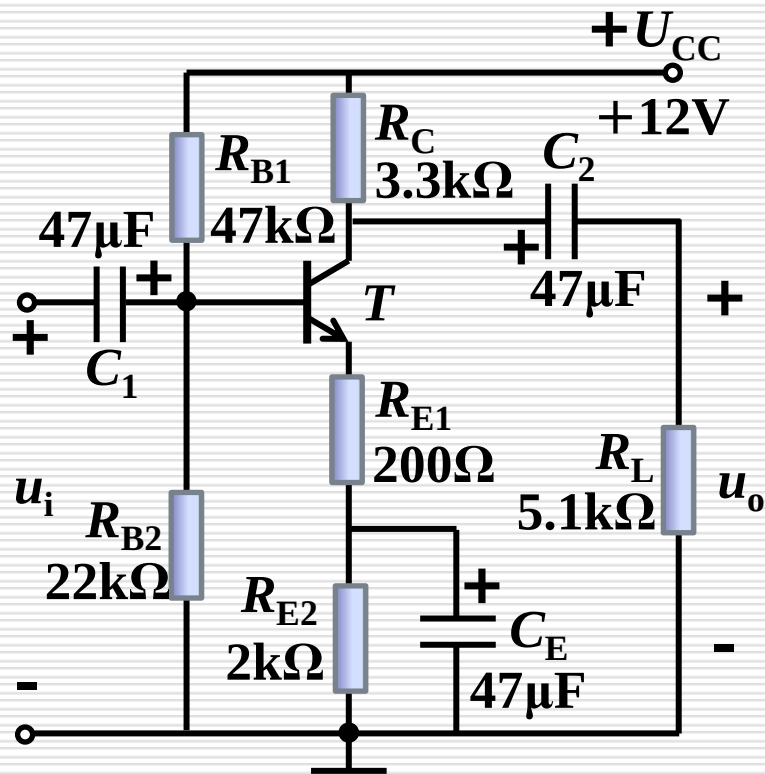
例题3.1.3

电路如右图所示,

$\beta=80$, $U_{BE}=0.7V$

试计算:

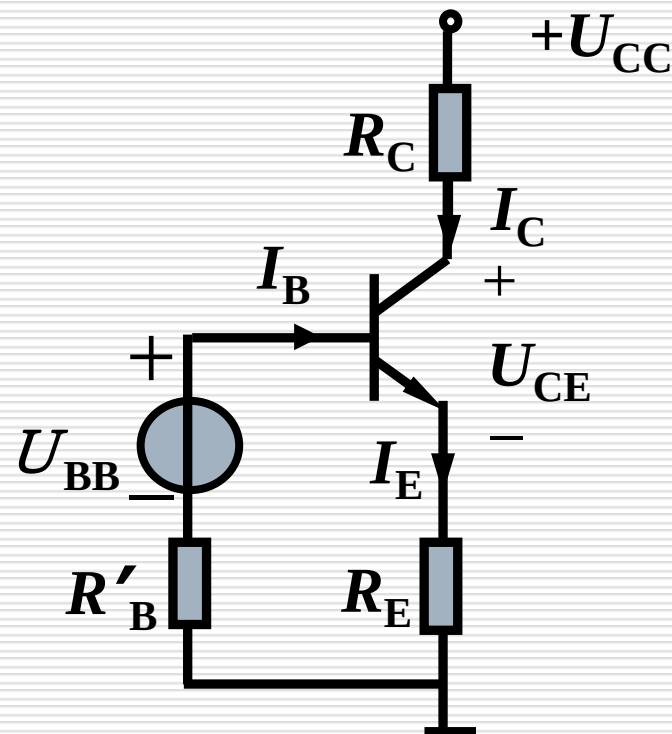
- (1) 电压放大倍数
- (2) 输入电阻
- (3) 输出电阻



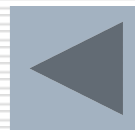
静态工作点:

$$U_{BB} = \frac{R_{B2}}{R_{B1} + R_{B2}} U_{CC}$$
$$= \frac{22 \times 10^3}{(47 + 22) \times 10^3} \times 12 = 3.83 \text{V}$$

$$R'_B = \frac{R_{B1} R_{B2}}{R_{B1} + R_{B2}} = \frac{47 \times 22}{47 + 22} = 14.99 \text{k}\Omega$$



直流等效电路



$$I_B = \frac{U_{BB} - U_{BE}}{R'_B + (1 + \beta) R_E} = \frac{3.83 - 0.7}{(14.99 + 81 \times 2.2) \times 10^3} = 0.0162(\text{mA})$$

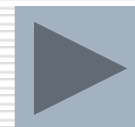
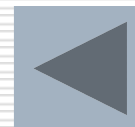
$$I_C = \beta I_B = 80 \times 0.0162 = 1.30(\text{mA})$$

$$I_E = (1 + \beta) I_B = 81 \times 0.0162 = 1.31(\text{mA})$$

$$U_{CE} = U_{CC} - R_C I_C - R_E I_E = 4.83(\text{V})$$

晶体管输入电阻

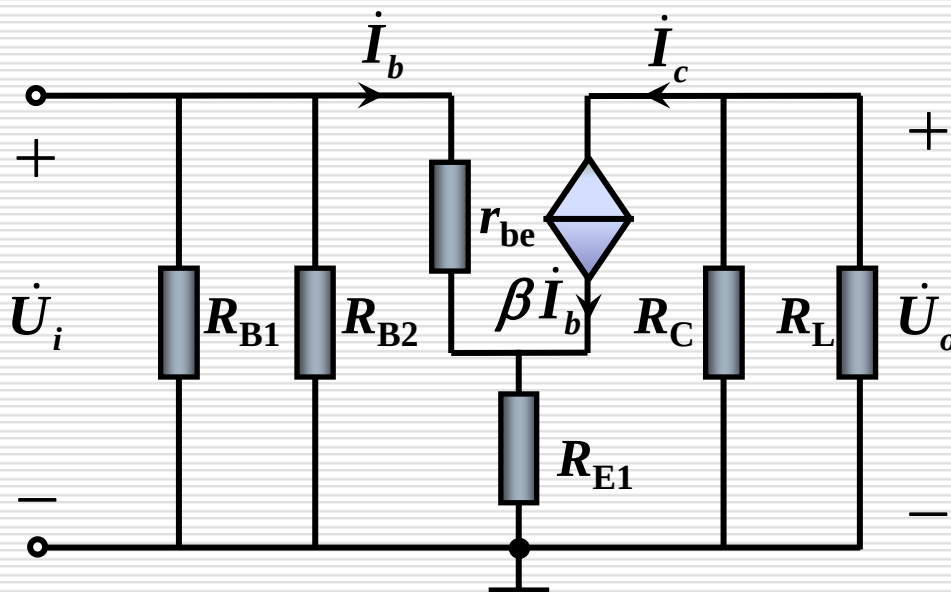
$$r_{be} = 200 + (1 + 80) \frac{26}{1.31} = 1.81(\text{k}\Omega)$$



微变等效电路：

$$R'_L = \frac{R_C R_L}{R_C + R_L}$$

$$= \frac{3.3 \times 5.1}{3.3 + 5.1} = 2\text{k}\Omega$$



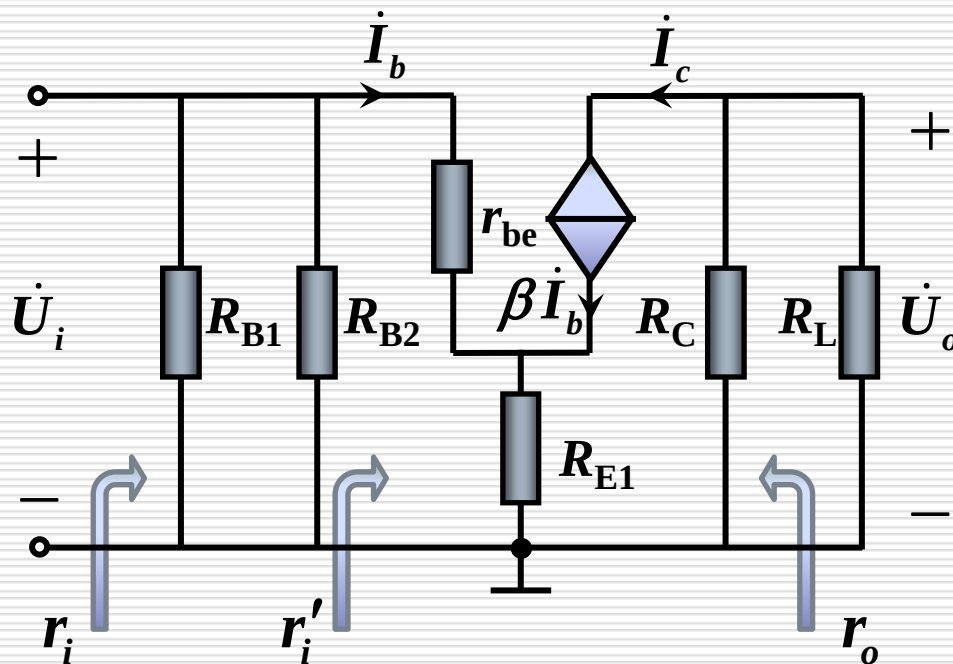
$$\dot{U}_i = [r_{be} + (1 + \beta)R_{E1}] \dot{I}_b \quad \dot{U}_o = -R'_L \dot{I}_c = -\beta R'_L \dot{I}_b$$

(1) 放大倍数

$$A_u = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta R'_L}{r_{be} + (1 + \beta)R_{E1}} = -\frac{80 \times 2}{1.81 + 81 \times 2} = -8.88$$



微变等效电路：



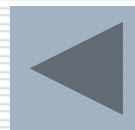
(2) 输入电阻

$$\dot{U}_i = [r_{be} + (1 + \beta)R_{E1}] \dot{I}_b$$

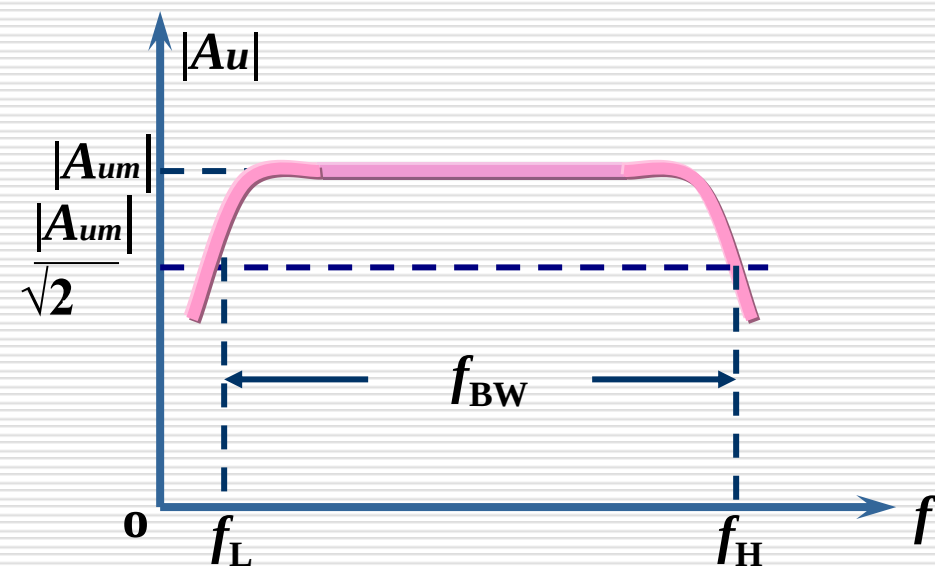
$$r'_i = \frac{\dot{U}_i}{\dot{I}_b} = r_{be} + (1 + \beta)R_{E1} = [1.81 + (1 + 80) \times 0.2] = 18.01(\text{k}\Omega)$$

$$r_i = R_{B1} // R_{B2} // r'_i = 8.18 \text{ k}\Omega$$

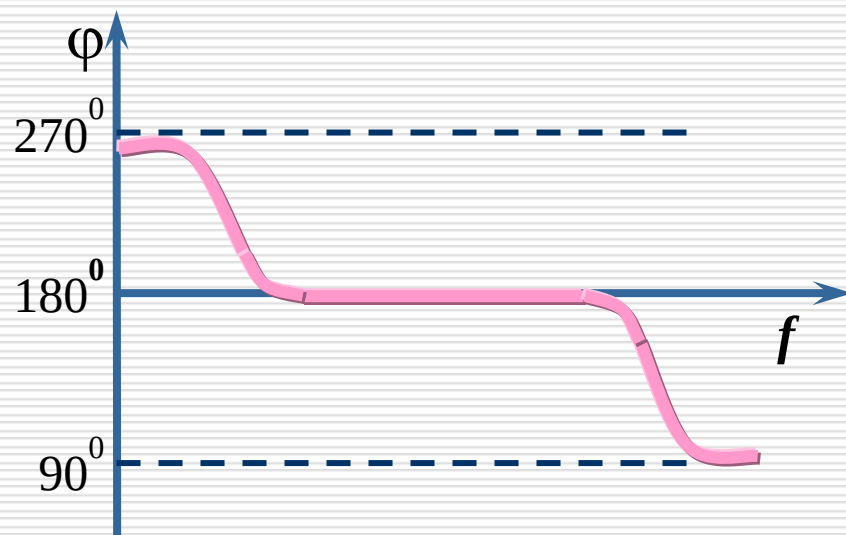
(3) 输出电阻 $r_o = R_C = 3.3 \text{ k}\Omega$



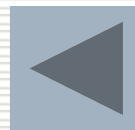
3.1.5 频率特性



幅频特性



相频特性



3.2 共集电极放大电路

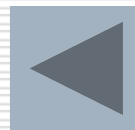
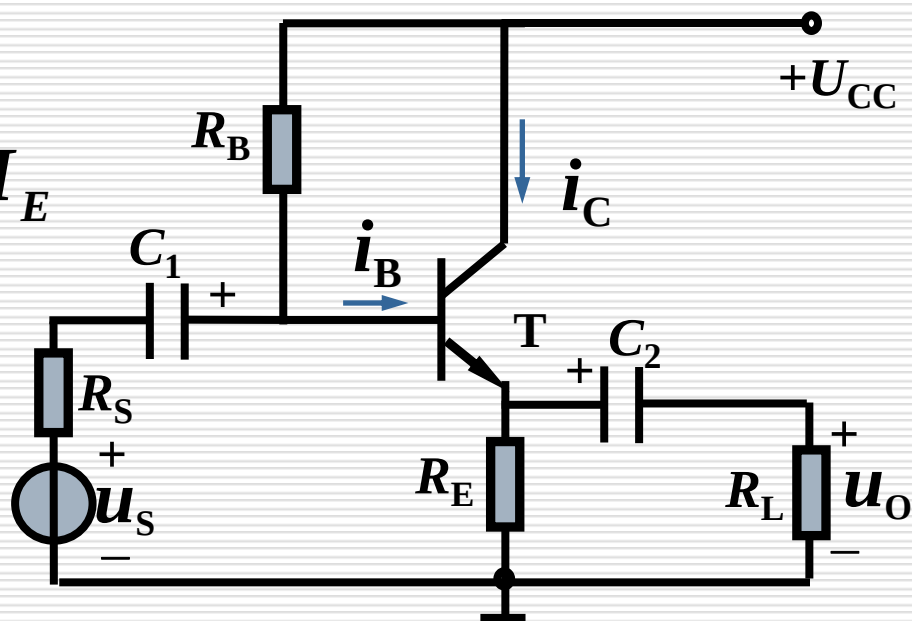
静态分析：

$$U_{CC} = R_B I_B + U_{BE} + R_E I_E$$

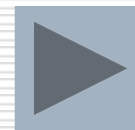
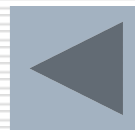
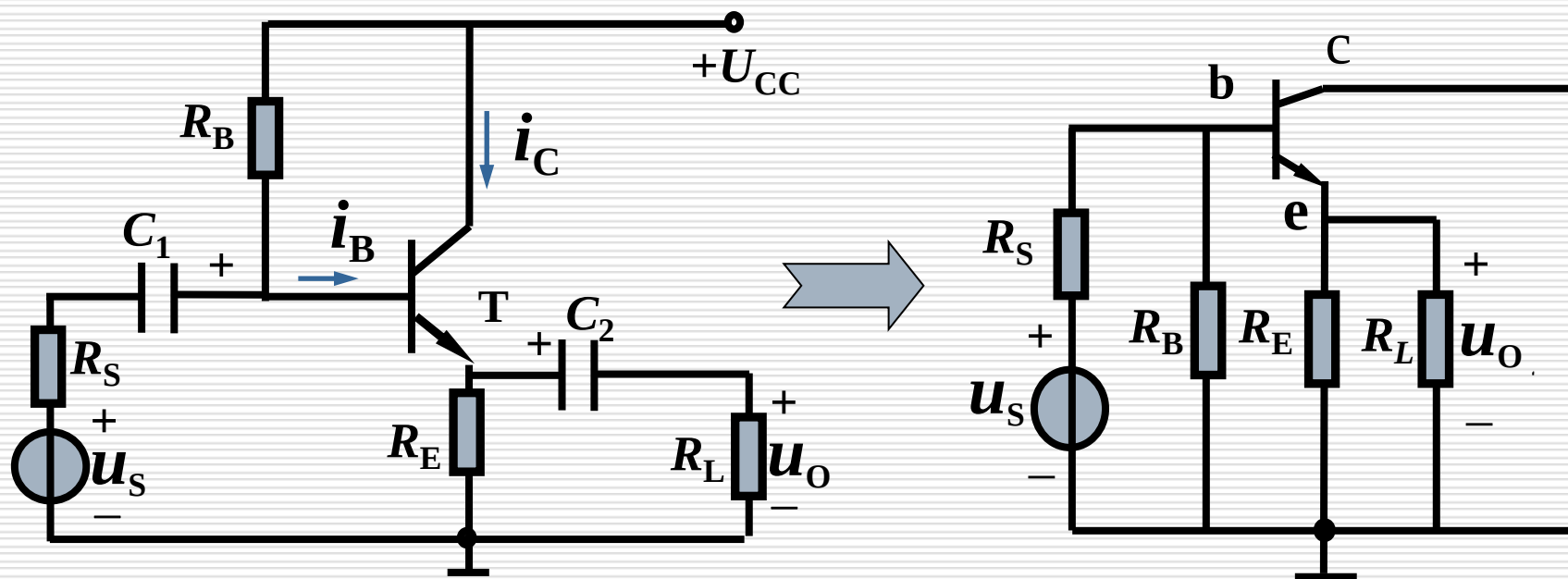
$$I_B = \frac{U_{CC} - U_{BE}}{R_B + (1 + \beta) R_E}$$

$$I_E = (1 + \beta) I_B$$

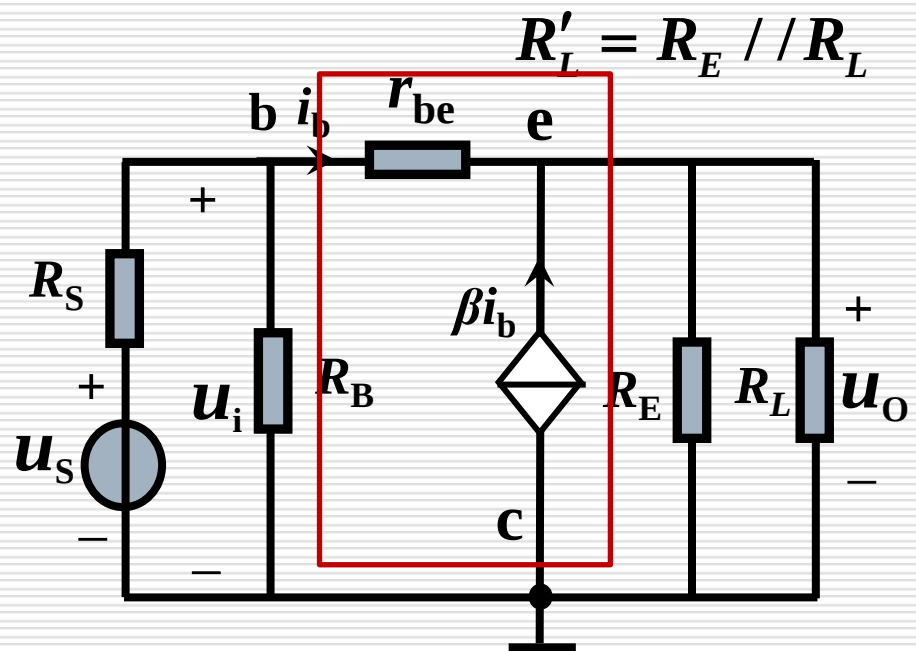
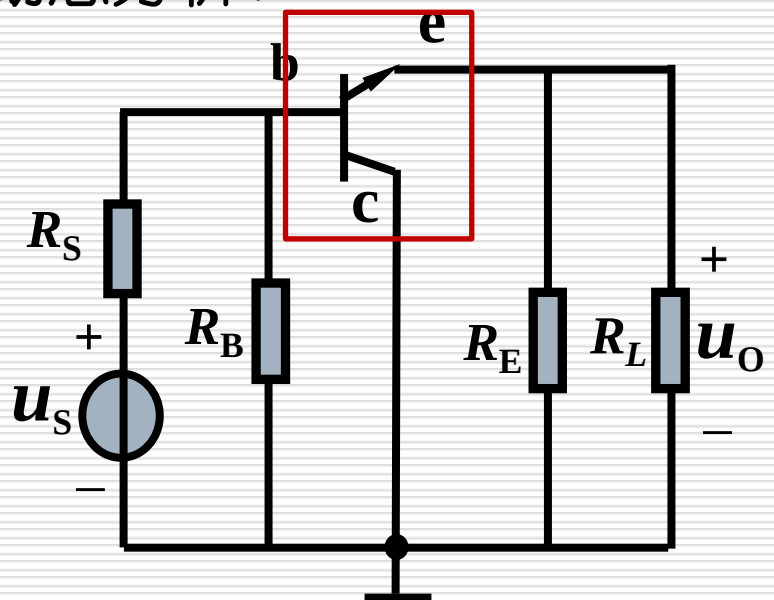
$$U_{CE} = U_{CC} - R_E I_E$$



共集电极放大电路交流通路



动态分析:



电压放大倍数 设小信号为正弦波，则电压电流可用相量表示

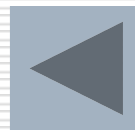
$$\dot{U}_i = r_{be} \dot{I}_b + (1 + \beta) R'_L \dot{I}_b$$

$$\dot{U}_o = (1 + \beta) R'_L \dot{I}_b$$

$$A_U = \frac{\dot{U}_o}{\dot{U}_i} = \frac{(1 + \beta) R'_L}{r_{be} + (1 + \beta) R'_L}$$

注意: (1) $A_U > 0$

(2) $A_U < 1$, $A_U \approx 1$



输入电阻

$$r_i = R_B // r'_i$$

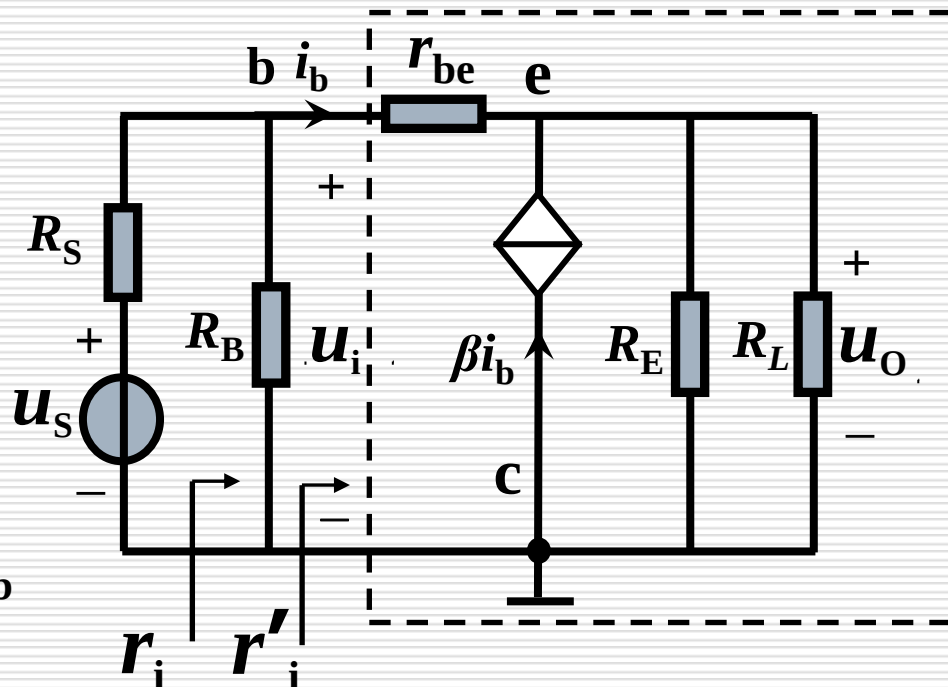
从

$$\dot{U}_i = r_{be} \dot{I}_b + (1 + \beta) R'_L \dot{I}_b$$

得

$$r'_i = \frac{\dot{U}_i}{\dot{I}_b} = r_{be} + (1 + \beta) R'_L$$

$$r_i = R_B // [r_{be} + (1 + \beta) R'_L]$$



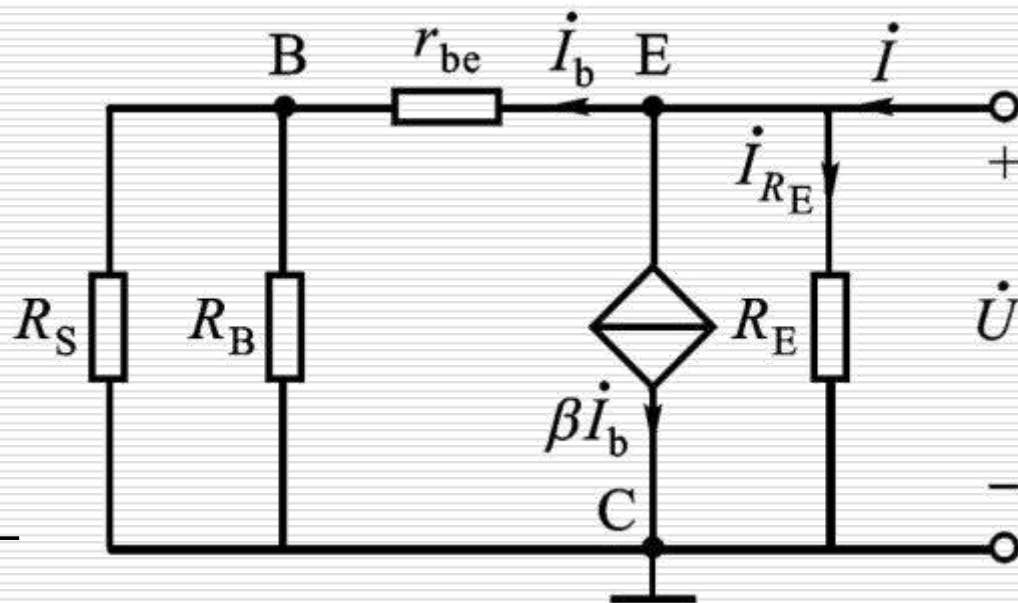
射极输出器输入电阻较大，说明对输入信号的衰减较少，常用于多级放大电路的输入级



——采用外施电压法

$$= \left(\frac{1}{R_E} + \frac{(1 + \beta)}{r_{be} + R'_S} \right) \dot{U}$$

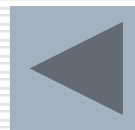
$$r_o = \frac{\dot{U}}{\dot{I}} = \frac{1}{\frac{1}{R_E} + \frac{1+\beta}{r_{be} + R'_S}} = R_E // \frac{r_{be} + R'_S}{1+\beta}$$



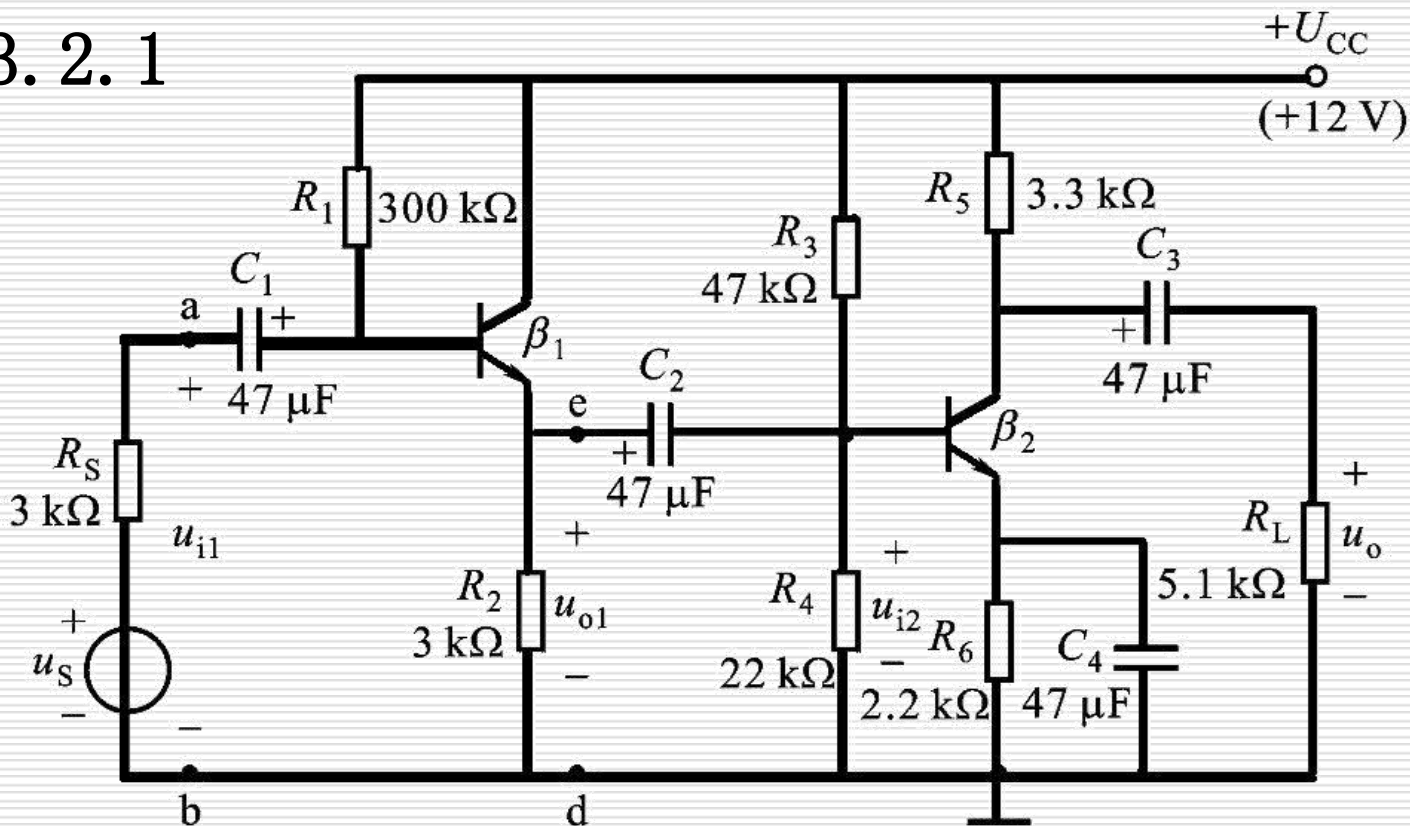
求 r_o 的等效电路

$$(R'_S = R_S // R_B)$$

射极输出器输出电阻较小，说明带负载能力较强，常用于小功率多级放大电路的输出级



例题3.2.1

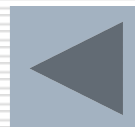


已知： $U_S = 20\text{mV}$ ， $U_{BE1} = U_{BE2} = 0.7\text{V}$ ， $\beta_1 = \beta_2 = 80$

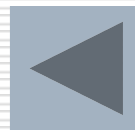
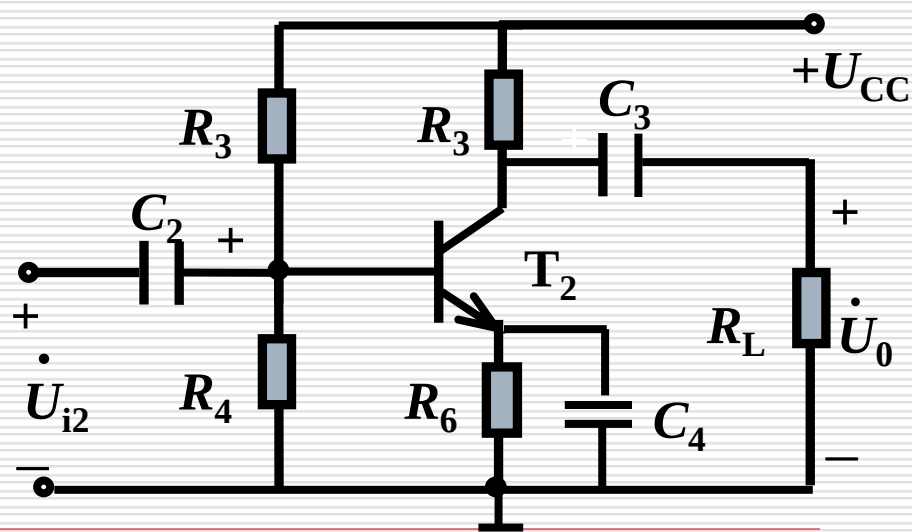
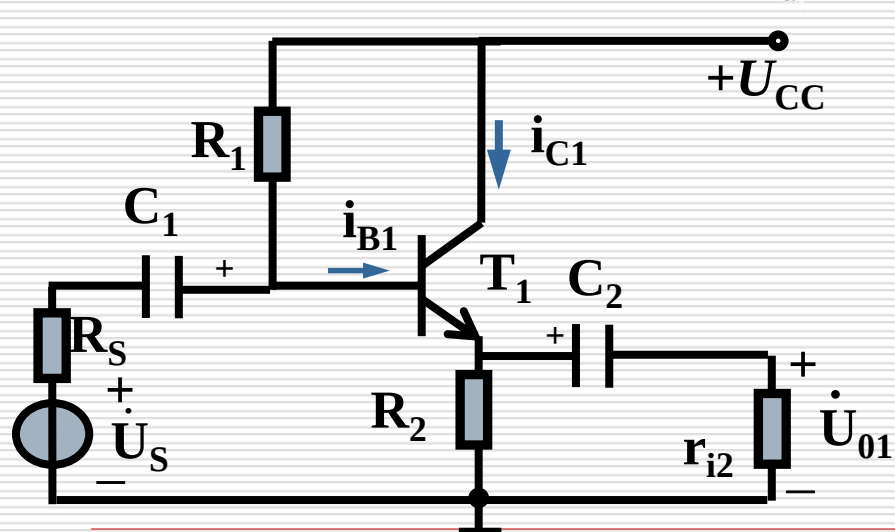
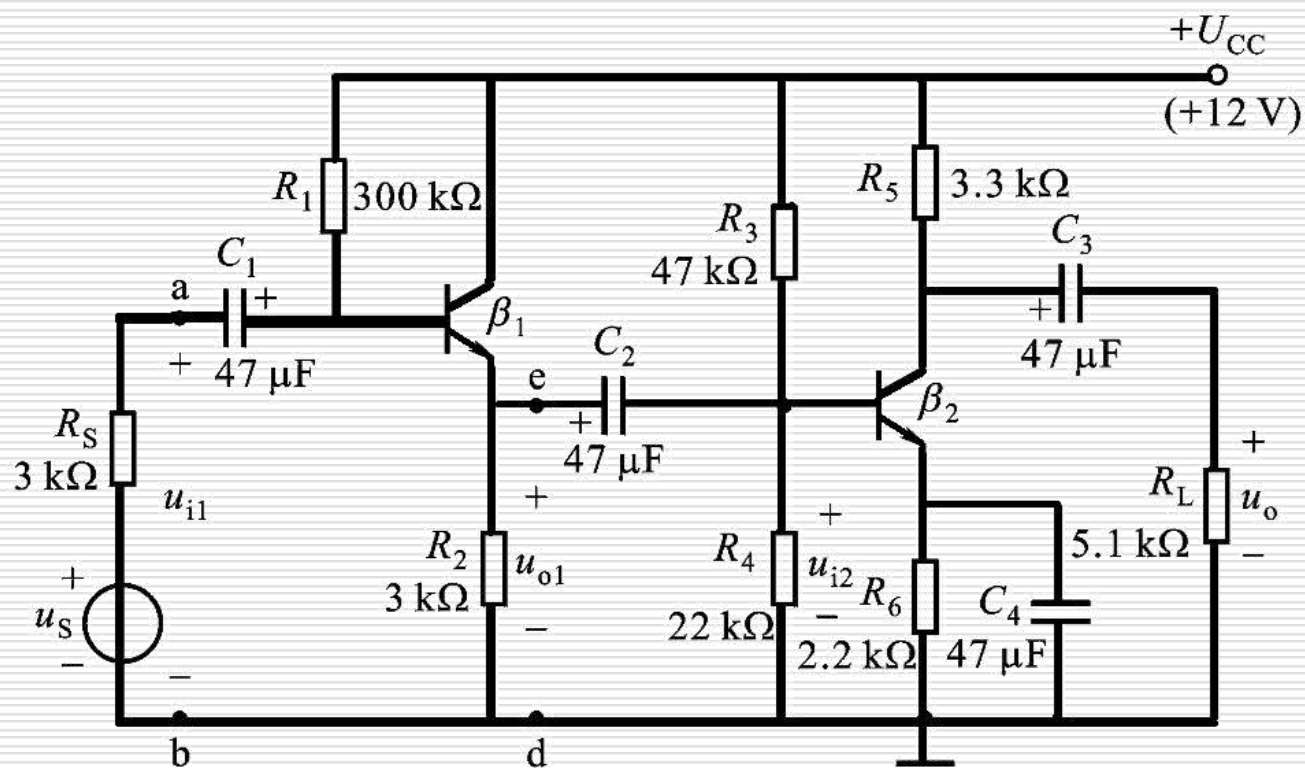
求： 1) 第一级 A_{u1} 、 r_{i1} 、 r_{o1} ；

2) U_{i2} ；

3) 若e、d与a、b直接相连， U_{i2} ？



解：



(1) 第一级静态电流

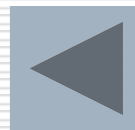
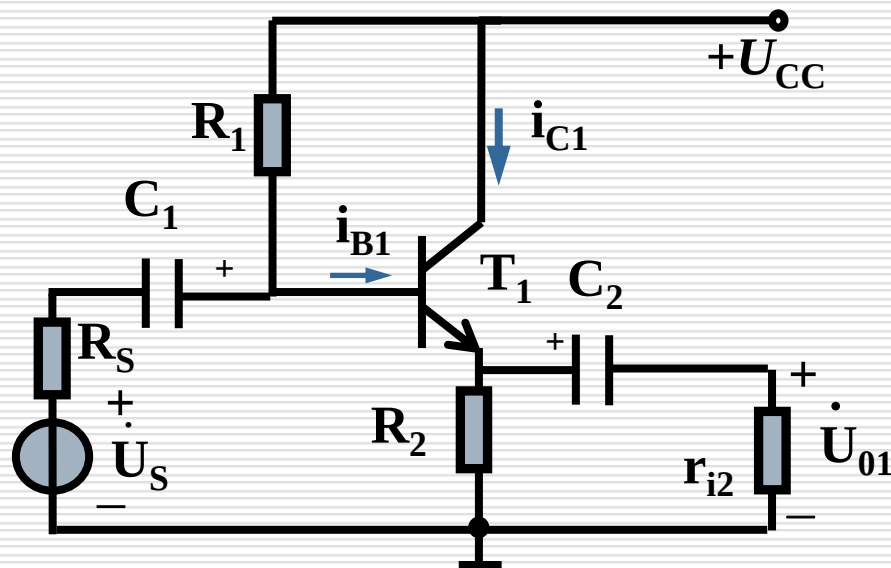
$$\begin{aligned} I_{B1} &= \frac{U_{CC} - U_{BE1}}{R_1 + (1 + \beta_1)R_2} \\ &= \frac{12 - 0.7}{(300 + 81 \times 3) \times 10^3} \\ &= 0.0208(\text{mA}) \end{aligned}$$

$$I_{E1} = (1 + \beta_1)I_{B1} = 81 \times 0.0208 = 1.68(\text{mA})$$

$$r_{be1} = 200 + (1 + \beta_1) \frac{26}{I_{E1}} = (200 + 81 \times \frac{26}{1.68}) = 1.45(\text{k}\Omega)$$

第二级的输入电阻 r_{i2} 即为第一级的负载电阻

$$R'_L = R_2 // r_{i2} = 1.05(\text{k}\Omega)$$



电压放大倍数

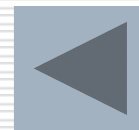
$$A_{u1} = \frac{(1 + \beta_1)R'_L}{r_{be1} + (1 + \beta_1)R'_L} = \frac{81 \times 1.05 \times 10^3}{(1.45 + 81 \times 1.05) \times 10^3} = 0.983$$

输入电阻

$$r_{i1} = R_1 // [r_{be1} + (1 + \beta_1)R'_L] = 67.1 \text{ k}\Omega$$

输出电阻

$$r_{o1} = R_2 // \frac{r_{be1} + R'_S}{1 + \beta_1} = 53.6 \text{ }\Omega$$



(2) 信号源输出电压

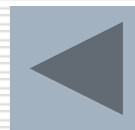
$$U_{i1} = \frac{r_{i1}}{R_s + r_{i1}} U_s = \frac{67.1 \times 10^3}{3 \times 10^3 + 67.1 \times 10^3} \times 20 \times 10^{-3} = 19.1(\text{mV})$$

第二级输入电压

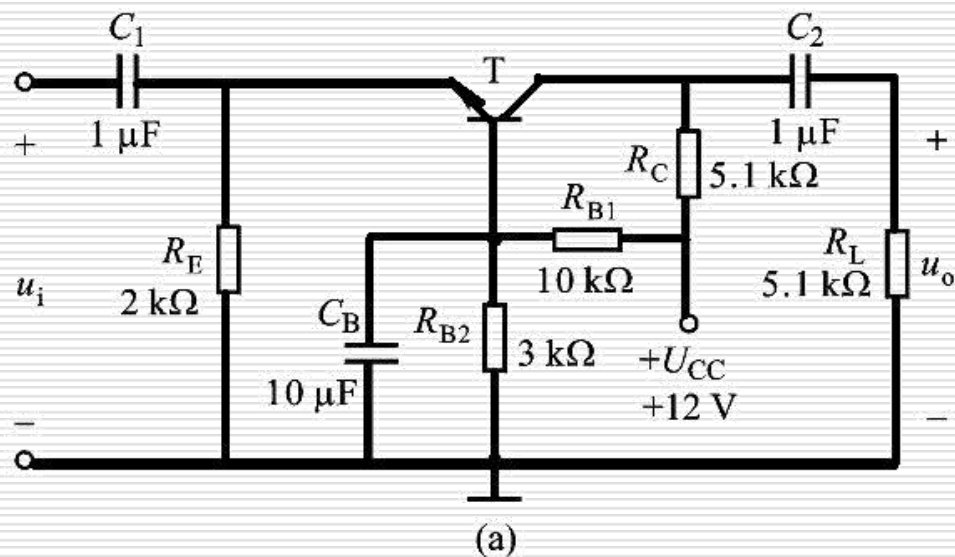
$$U_{i2} = U_{o1} = |A_{u1}| \times U_{i1} = 0.983 \times 19.1 = 18.8(\text{mV})$$

(3) 若e、d与a、b直接相连, 则 U_{i2} 就是信号源的输出电压

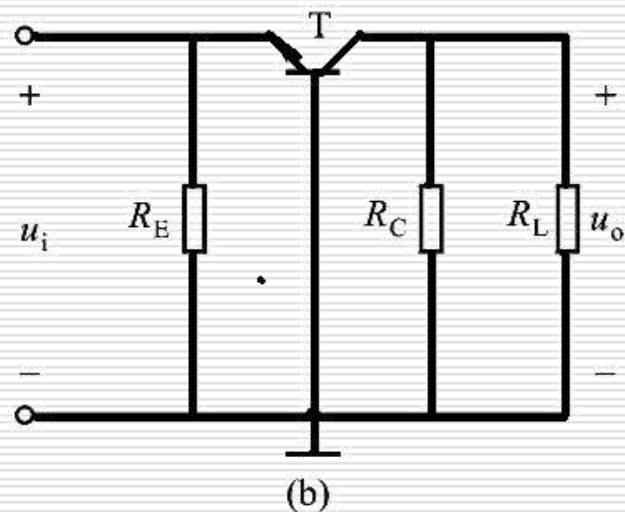
$$U_{i2} = \frac{r_{i2}}{R_s + r_{i2}} U_s = \frac{1.61 \times 10^3}{(3 + 1.61) \times 10^3} \times 20 \times 10^{-3} = 6.89(\text{mV})$$



共基极放大电路

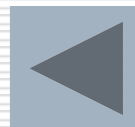


基本放大电路



交流通路

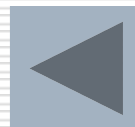
电路只能放大电压信号，不能放大电流；
输出与输入同相， r_i 小， r_o 大。



3.3 共源极放大电路

3.3.1 静态分析

3.3.2 动态分析



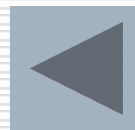
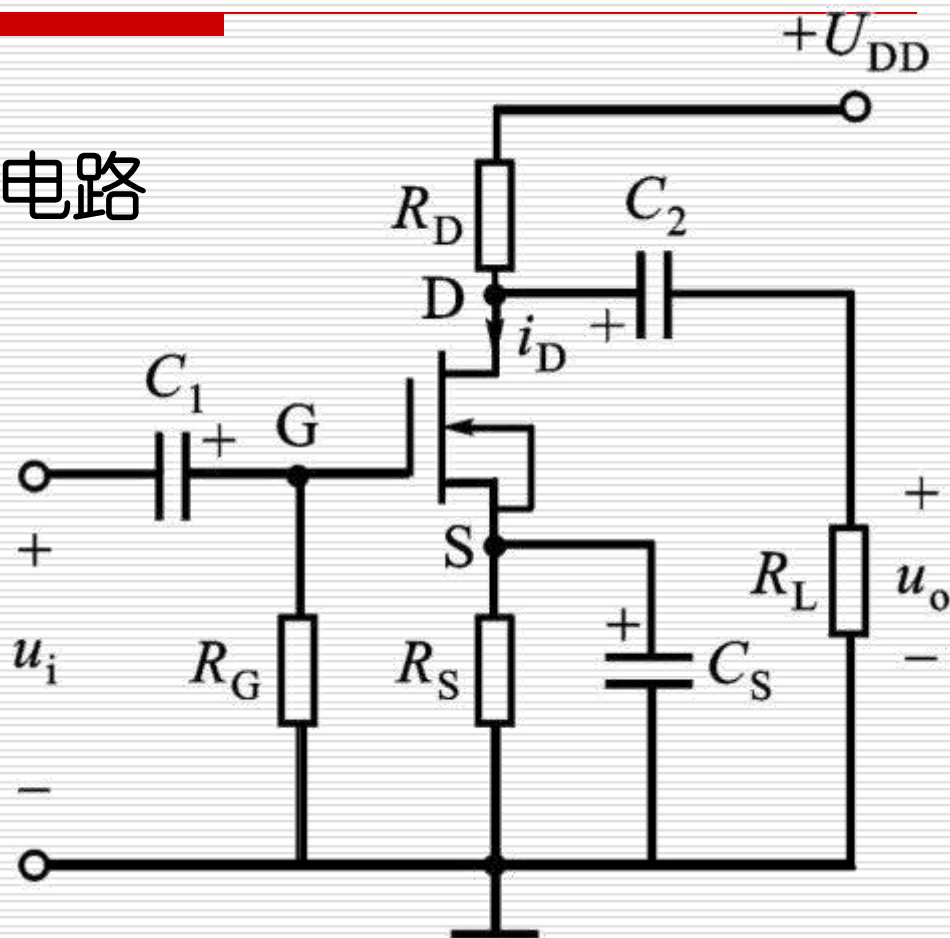
3.3.1 静态分析

自给式偏置共源极放大电路

$$\because U_G = 0$$

$$\therefore U_{GS} = -R_S I_D$$

仅适用于耗尽型场
效应管

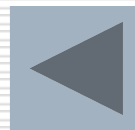
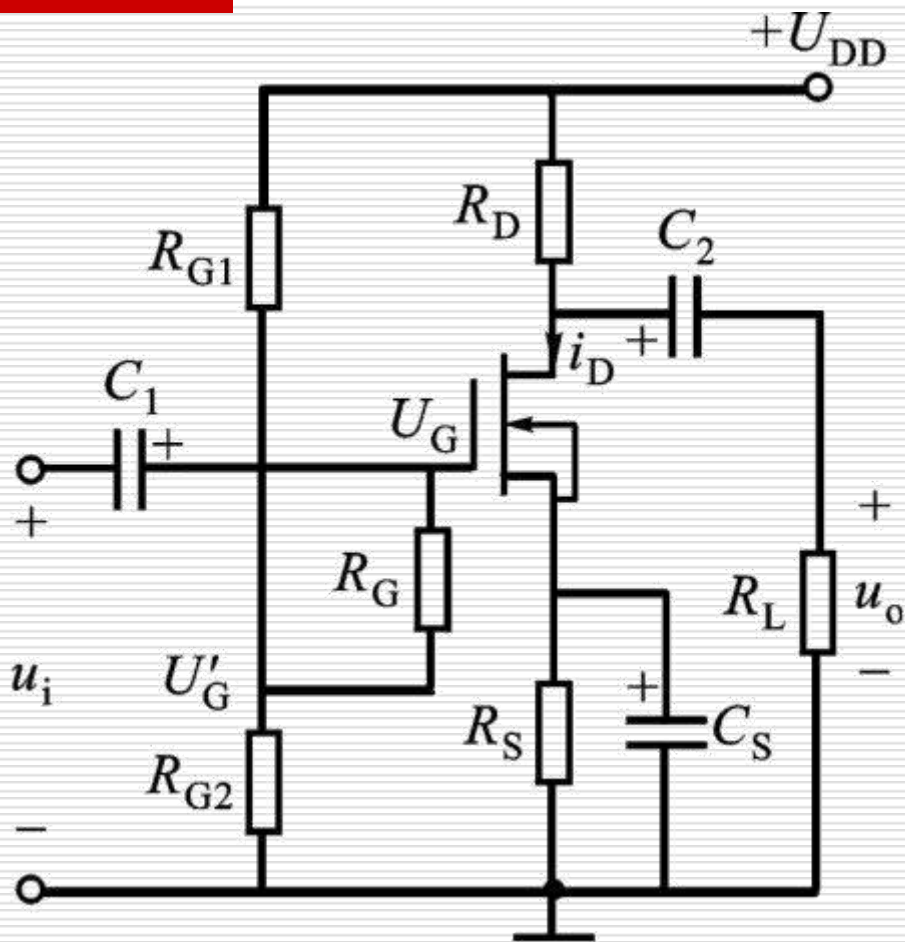


分压式自偏置共源极放大电路

$$U_G = U'_G = \frac{R_{G2}}{R_{G1} + R_{G2}} U_{DD}$$
$$\begin{cases} U_{GS} = U_G - U_S \\ = \frac{R_{G2}}{R_{G1} + R_{G2}} U_{DD} - R_S I_D \\ I_D = I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS(off)}}\right)^2 \end{cases}$$

(耗尽型)

两方程联立求得 I_D 和 U_{GS}



3.3.2 动态分析

电压放大倍数

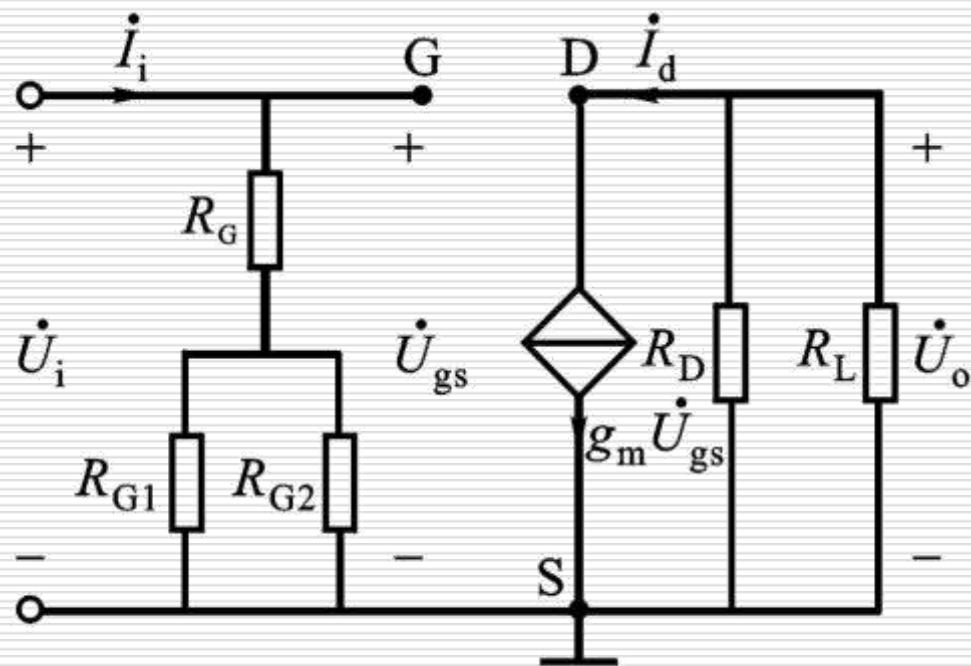
$$\dot{A}_U = \frac{\dot{U}_o}{\dot{U}_i} = \frac{-g_m R'_L \dot{U}_{gs}}{\dot{U}_{gs}} = -g_m R'_L$$

输入电阻

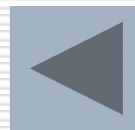
$$r_i = \frac{\dot{U}_i}{\dot{I}_i} = R_G + \frac{R_{G1} R_{G2}}{R_{G1} + R_{G2}}$$

输出电阻

$$r_o = R_D$$



微变等效电路



【例题3.3.1】 已知电路如图，且 $g_m = 2\text{mS}$ ， $U_{GS(\text{off})} = -2\text{V}$ ， $I_{DSS} = 0.5\text{mA}$ 。

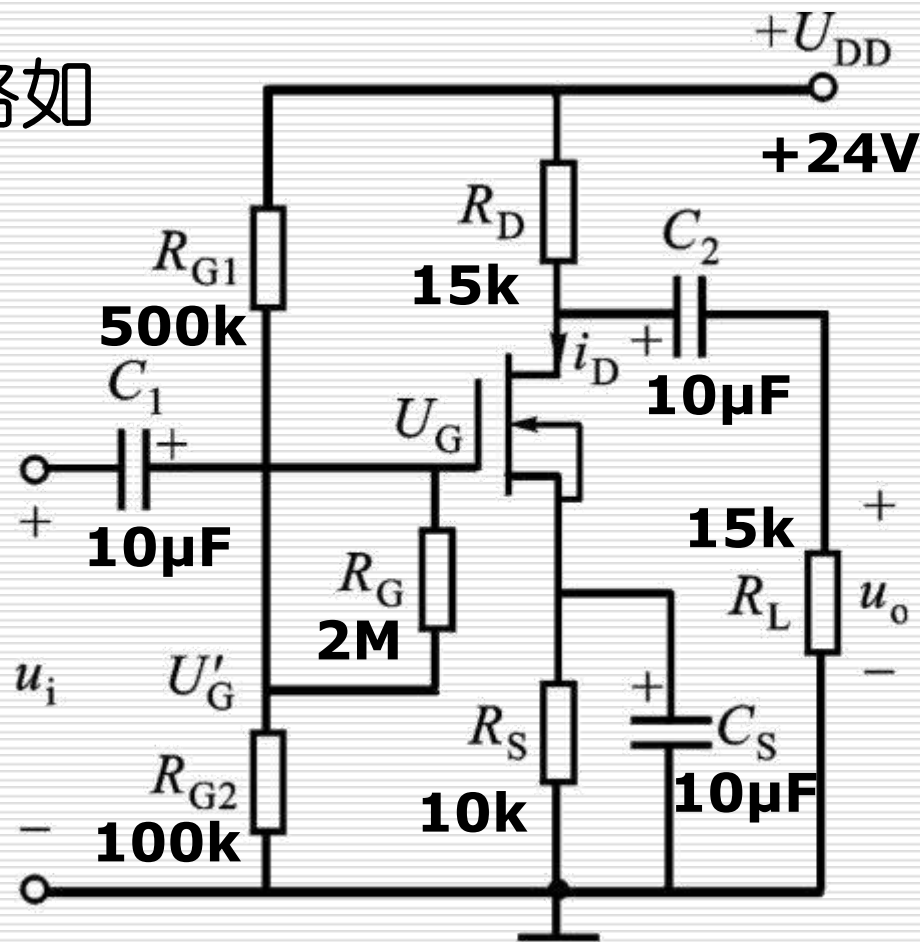
求：(1) 静态工作点 I_D 、 U_{GS} 、 U_{DS}

(2) A_U 、 r_i 、 r_o 。

【解】 (1)

$$U_G = \frac{R_{G2}}{R_{G1} + R_{G2}} U_{DD}$$

$$= \frac{100}{(500 + 100)} \times 24 = 4\text{V}$$



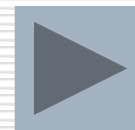
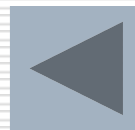
$$\left. \begin{aligned} U_{GS} &= U_G - R_S I_D = 4 - 10I_D \\ I_D &= I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS(off)}} \right)^2 = 0.5 \times \left(1 + \frac{U_{GS}}{2} \right)^2 \end{aligned} \right\} \text{联立解得}$$

$$\left. \begin{aligned} I_{D1} &= 0.863\text{mA} \\ U_{GS1} &= -4.63\text{V} \end{aligned} \right\} \text{①}$$

$$\left. \begin{aligned} I_{D2} &= 0.417\text{mA} \\ U_{GS1} &= -0.17\text{V} \end{aligned} \right\} \text{②}$$

根据题意第①组解不符，因此静态工作点为解②

$$\begin{aligned} U_{DS} &= U_{DD} - (R_D + R_S)I_D \\ &= 24 - (15 + 10) \times 0.417 = 13.6\text{V} \end{aligned}$$



(2) 电压放大倍数

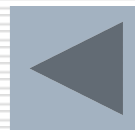
$$A_U = -g_m R_L' = -15$$

输入电阻

$$r_i = R_G + \frac{R_{G1} R_{G2}}{R_{G1} + R_{G2}} = 2.08 \text{M}\Omega$$

输出电阻

$$r_o = R_D = 15 \text{k}\Omega$$

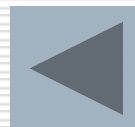


3.4 分立元件组成的基本门电路

3.4.1 二极管与门电路

3.4.2 二极管或门电路

3.4.3 晶体管及场效晶体管非门电路



基本门电路概述

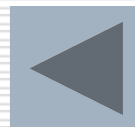
门电路是一种**开关电路**，这种电路的输入和输出之间存在一定的因果关系即逻辑关系。

在逻辑电路中，输入输出信号通常用高低电平来描述，用0和1来表示两种对立的逻辑状态。

正逻辑：1表示高电平，0表示低电平；

负逻辑：0表示高电平，1表示低电平。

基本的逻辑关系有：与逻辑、或逻辑、非逻辑；
相对应的基本门电路有：与门、或门、非门。



3.4.1 二极管与门电路

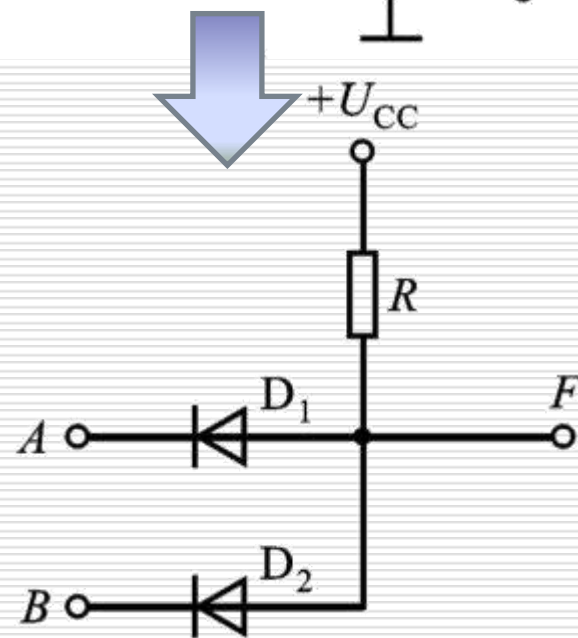
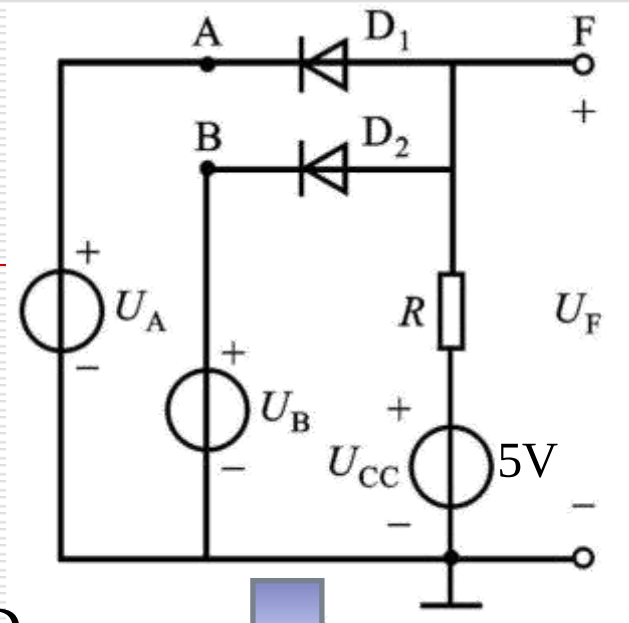
工作原理：

(1) 当 $U_A=U_B=0V$ 时， D_1 、 D_2 均导通，故 $U_F=0V$ ，即F为低电平0；

(2) 当 $U_A=0V$ ， $U_B=3V$ 时， D_1 导通， D_2 截止，故 $U_F=0V$ ，即F为低电平0；

(3) 当 $U_A=3V$ ， $U_B=0V$ 时， D_1 截止， D_2 导通，故 $U_F=0V$ ，即F为低电平0；

(4) 当 $U_A=U_B=3V$ 时， D_1 、 D_2 均导通，故 $U_F=3V$ ，即F为高电平1。

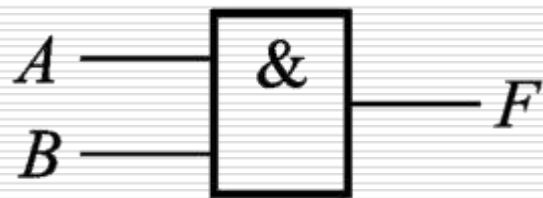


3.4.1 二极管与门电路

二输入与门逻辑表

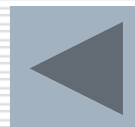
输 入		输 出
A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

二输入与门逻辑符号



函数表达式：

$$F = A \cdot B$$

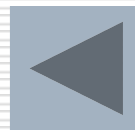
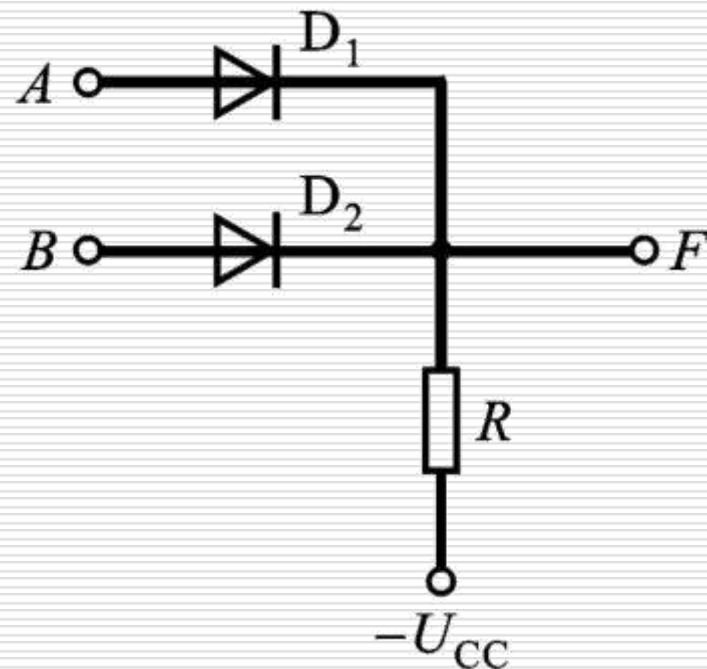


3. 4. 2 二极管或门电路

工作原理：

(1) 当 $U_A=U_B=0V$ 时， D_1 、 D_2 均导通，故 $U_F=0V$ ，即F为低电平0；

(2) 当 U_A 、 U_B 不全为0时， D_1 、 D_2 至少有一个导通。如 $U_A=3V$ 时， D_1 导通，此时 $U_F=U_A=3V$ ，即F为高电平1。

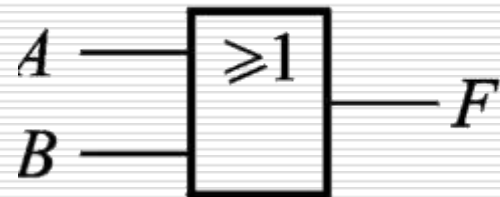


3.4.2 二极管或门电路

二输入或门逻辑表

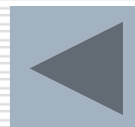
输 入		输 出
A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

二输入或门逻辑符号



函数表达式：

$$F = A + B$$



3.4.3 晶体管及场效晶体管非门电路

1. 晶体管非门电路

输入 U_A 为低电平0时, $U_{BE} < 0$,
晶体管发射结反偏, 管子截止,
输出端F为高电平1;

输入 U_A 为高电平时, 管子饱和,
输出端F为低电平0。

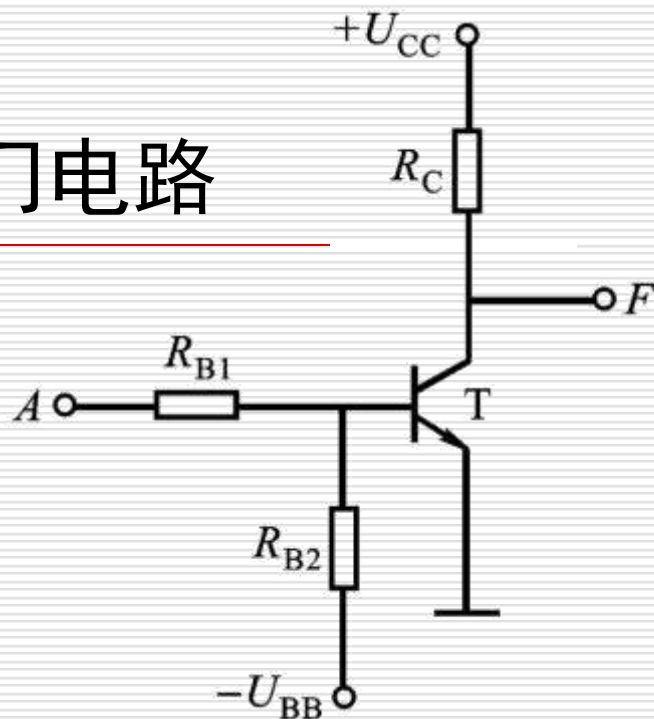
要使得管子饱和, 须满足

$$I_{CS} = \frac{U_{CC} - U_{CES}}{R_C} \approx \frac{U_{CC}}{R_C}$$

此时基极电流至少为 $I_{BS} = \frac{I_{CS}}{\beta}$ —— 临界饱和基极电流

保证晶体管饱和的条件

$$I_B \geq I_{BS}$$

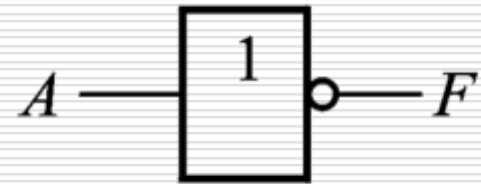


非门逻辑表、逻辑符号及函数表达式

非门逻辑表

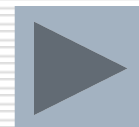
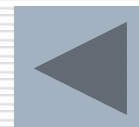
输 入	输 出
A	F
0	1
1	0

非门逻辑符号



函数表达式：

$$F = \overline{A}$$



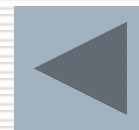
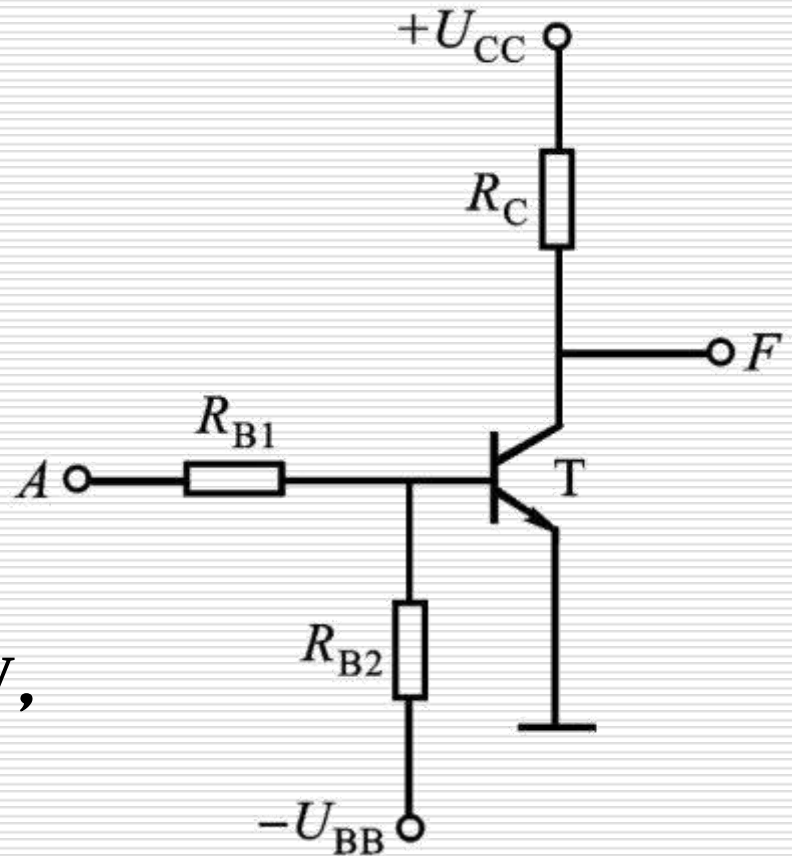
例3.4.1

已知： $R_{B1}=2.7\text{k}\Omega$, $R_{B2}=10\text{k}\Omega$, $R_C=1\text{k}\Omega$, $U_{CC}=5\text{V}$, $-U_{BB}=-5\text{V}$,
 $\beta=30$, $U_{BES}=0.7\text{V}$, $U_{CES}\approx 0\text{V}$, 截止时 $I_C\approx 0$, $U_{IL}=0\text{V}$,
 $U_{IH}=3\text{V}$ ；试分析管子的输出状态。

解：（1）输入 U_A 为低电平0时

$$U_{BE} = -\frac{R_{B1}}{R_{B1} + R_{B2}} U_{BB}$$
$$= -\frac{2.7 \times 10^3}{(2.7 + 10) \times 10^3} \times 5 = -1.06(\text{V})$$

此时晶体管可靠截止， $U_F \approx U_C = 5\text{V}$ ，
输出F为高电平。



(2) 输入 U_A 为高电平1时
 将晶体管基极左侧作戴维宁等效 开路电压

$$U_{BO} = U_{IH} - \frac{U_{IH} + U_{BB}}{R_{B1} + R_{B2}} R_{B1}$$

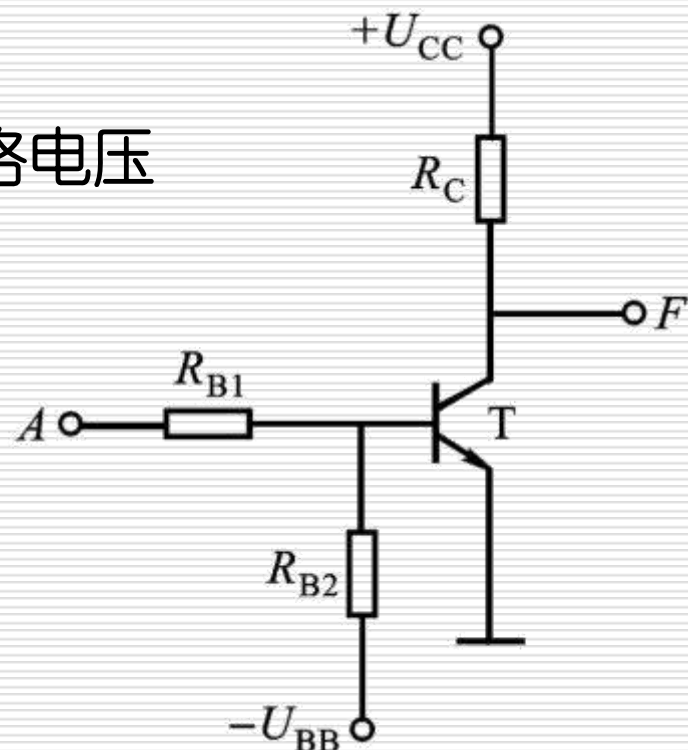
$$= \left[3 - \frac{3 + 5}{(2.7 + 10) \times 10^3} \times 2.7 \times 10^3 \right] = 1.3(\text{V})$$

等效电阻 $R_B = R_{B1} // R_{B2} = 2.13\text{k}\Omega$

基极电流 $I_B = \frac{U_{BO} - U_{BES}}{R_B} = \frac{1.3 - 0.7}{2.13 \times 10^3} = 0.28(\text{mA})$

临界基极饱和电流 $I_{BS} = \frac{I_{CS}}{\beta} = \frac{U_{CC}}{\beta R_C} = \frac{5}{30 \times 1 \times 10^3} = 0.17(\text{mA})$

因 $I_B > I_{BS}$, 晶体管处于饱和, $U_F = U_{CES} \approx 0\text{V}$, F为0。



2. 场效晶体管非门电路

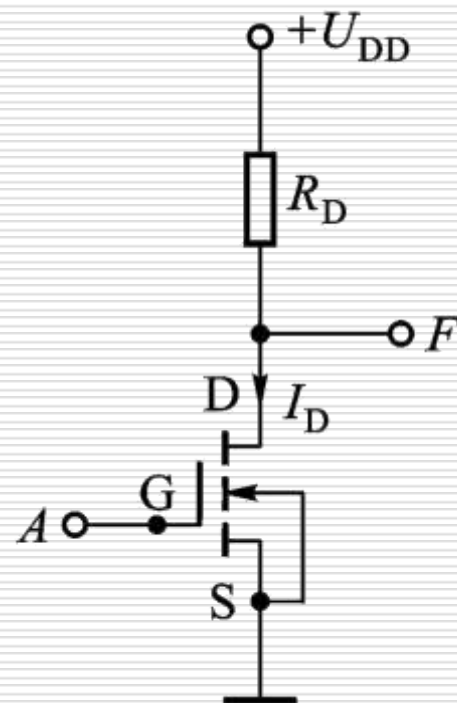
NMOS非门电路

工作原理：

(1) 当输入 U_A 为低电平0时，使 $U_{GS} < U_{GS(th)}$ ，管子处于截止状态， $I_D = 0, U_F = U_{DD}$ ；输出端F为高电平1；

(2) 当输入 U_A 为高电平1时， $U_{GS} > U_{GS(th)}$ ，管子饱和导通， $U_F \approx 0$ ，输出端F为低电平0。

故电路满足非逻辑关系。



互补型MOS非门电路

T_N : 增强型NMOS管, 驱动管; T_P : 增强型PMOS管, T_N 的负载管。

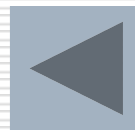
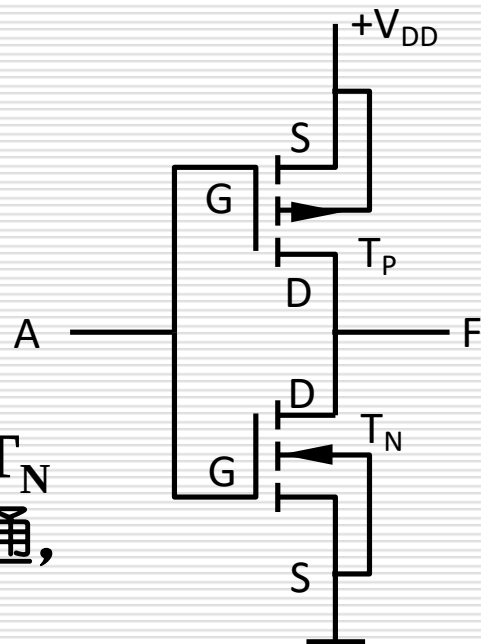
设 T_N 和 T_P 的开启电压为 $U_{GSN(th)}$ 和 $U_{GSP(th)}$

(1) $A=0$ (设 $u_A=0V$), $U_{GSN}=0 < U_{GSN(th)}$, T_N 截止, 而 $|U_{GSP}| \approx U_{DD} > |U_{GSP(th)}|$, T_P 导通, $u_F \approx U_{DD}$, $F=1$, 为高电平。

(2) $A=1$ (设 $u_A=U_{DD}$), $U_{GSN}=U_{DD} > U_{GSN(th)}$, T_N 导通, 而 $U_{GSP} \approx 0V < |U_{GSP(th)}|$, T_P 截止, $u_F \approx 0V$, $F=0$, 为低电平。

从以上分析可知, 该电路具有非门的功能。

相比于NMOS非门电路, 互补型MOS在工作时只有一个MOS管导通, 静态功耗很小; 另外因 T_N 、 T_P 导通电阻小, 负载电容充电或放电很快, 故工作速度更快。



本章结束

[返回目录](#)

第4章 数字集成电路